# Supplement S1: Post-review controls and uncertainty

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## 1 Status and scope

This supplement supersedes broad interpretations of conditional component gains in earlier S2–S4 materials. All added analyses reuse exposed R2 outcomes and are exploratory. No new inference or participants were used. All 658 TEST participants and 49 endpoints are complete; CAL contains 200 people. Source and protocol hashes were verified before analysis. Recomputed original Granite CAL weights and selected TEST losses agree to $10^{-12}$. Individual forecasts and responses remain restricted. Aggregate JSON supports arithmetic checks but cannot independently reproduce participant bootstrap intervals.

## 2 Controls

The original decoder1200 is fixed. Each control linearly pools it with (1) the FIT-1000 Laplace endpoint prior, (2) uniform probabilities or (3) a FIT empirical Laplace prior within exact question-text-and-options cells, falling back to (1) if unseen. The frozen grid is $0, .1, \ldots, 1$. Each endpoint chooses the lowest CAL native proper loss; ties choose the smallest control weight. The statistical library selects jointly among these three candidates and eleven weights, with ties resolved by weight then listed candidate order. Its 33 candidates exceed the eleven-candidate individual mixture search; it is not an equal-search-budget comparison. We did not fit a new learned-model library. The stimulus-only control has the same eleven-weight budget as each LLM mixture.

The person-invariant proxies average Granite CAL forecasts within these same stimulus cells separately for neutral and T0. CAL evaluation leaves out the current person; TEST uses CAL means only. There were zero empty-cell fallbacks. They test a pooling surrogate, not misassigned-history inference; forecast dispersion also changes. No TEST outcomes or forecasts enter proxy construction.

Conditional paired percentile intervals resample 658 TEST participants jointly across endpoints, 10,000 draws, seed 20261110. They retain selected weights and frozen endpoint normalization. There is no new multiplicity-corrected success declaration. Equal-family risk weights the forty economic endpoints collectively by one half and the nine heuristic endpoints by one half. Native losses below are kept separate.

## 3 All absolute risks

| Predictor | Normalized | Economic LL | Ordinal RPS | Other LL |
|---|---|---|---|---|
| decoder1200 | 0.910600 | 0.533033 | 0.118356 | 0.662077 |
| full1400 | 0.911749 | 0.532547 | 0.118963 | 0.661801 |
| G neutral | 1.344421 | 0.684230 | 0.198466 | 0.728581 |
| G t0 | 1.287389 | 0.674690 | 0.183113 | 0.850429 |
| qwen neutral | 1.220471 | 0.745435 | 0.155558 | 0.680865 |
| qwen t0 | 1.251136 | 0.756636 | 0.159960 | 0.678840 |
| G neutral fixed half | 1.007893 | 0.559952 | 0.137646 | 0.681000 |
| G neutral mix | 0.907714 | 0.531121 | 0.118017 | 0.662077 |
| G t0 fixed half | 0.992903 | 0.556917 | 0.133776 | 0.710476 |
| G t0 mix | 0.907911 | 0.531498 | 0.117983 | 0.662077 |
| qwen neutral fixed half | 0.980769 | 0.580178 | 0.126841 | 0.665214 |
| qwen neutral mix | 0.909728 | 0.532049 | 0.118306 | 0.665214 |
| qwen t0 fixed half | 0.988393 | 0.581649 | 0.128169 | 0.665576 |
| qwen t0 mix | 0.910650 | 0.532114 | 0.118457 | 0.665576 |

| | | | | |
|---|---|---|---|---|
| endpoint-prior mix | 0.909403 | 0.532989 | 0.118007 | 0.662006 |
| uniform mix | 0.910043 | 0.532592 | 0.118273 | 0.662077 |
| stimulus-prior mix | 0.907341 | 0.530756 | 0.118007 | 0.662006 |
| neutral proxy | 0.909798 | 0.532193 | 0.118336 | 0.662077 |
| t0 proxy | 0.909421 | 0.532117 | 0.118244 | 0.662077 |
| statistical library mix | 0.907577 | 0.531078 | 0.118007 | 0.662006 |
| FIT prior (unmixed) | 1.000056 | 0.640051 | 0.119758 | 0.662006 |

# 4 Fourteen exploratory contrasts

| System | Reference | Difference | Conditional 95% CI |
|---|---|---|---|
| G neutral mix | decoder1200 | -0.002886 | [-0.004478, -0.001282] |
| G neutral mix | full1400 | -0.004036 | [-0.007178, -0.000932] |
| G neutral mix | endpoint-prior mix | -0.001690 | [-0.003562, +0.000214] |
| G neutral mix | uniform mix | -0.002329 | [-0.003837, -0.000824] |
| G neutral mix | stimulus-prior mix | +0.000373 | [-0.001864, +0.002570] |
| G neutral mix | statistical library mix | +0.000136 | [-0.002094, +0.002351] |
| G neutral mix | neutral proxy | -0.002084 | [-0.002991, -0.001156] |
| G t0 mix | decoder1200 | -0.002689 | [-0.004239, -0.001123] |
| G t0 mix | full1400 | -0.003839 | [-0.006961, -0.000756] |
| G t0 mix | endpoint-prior mix | -0.001493 | [-0.003472, +0.000476] |
| G t0 mix | uniform mix | -0.002132 | [-0.003549, -0.000691] |
| G t0 mix | stimulus-prior mix | +0.000570 | [-0.001706, +0.002783] |
| G t0 mix | statistical library mix | +0.000333 | [-0.001919, +0.002562] |
| G t0 mix | t0 proxy | -0.001510 | [-0.002554, -0.000472] |

# 5 CAL and TEST resampling

We independently resample 200 CAL and 658 TEST people with replacement, preserving all endpoints within each participant. For each of 2,000 draws (seed 20261111), CAL losses reselect each endpoint weight using the original grid and tie rule. The TEST draw scores that selection. Model fits, instruction forecasts and normalization denominators remain frozen: these are descriptive selection-sensitivity intervals, not fully nested training uncertainty. Resampling a discrete minimizer can be nonregular; no coverage guarantee is claimed. Full1400 is also held fixed, despite having used the original CAL labels for fitting.

| System | Reference | Bootstrap mean | Percentile interval |
|---|---|---|---|
| neutral | decoder1200 | -0.001901 | [-0.004268, +0.000902] |
| neutral | full1400 | -0.003055 | [-0.006759, +0.000864] |
| t0 | decoder1200 | -0.001384 | [-0.003744, +0.001577] |
| t0 | full1400 | -0.002539 | [-0.006185, +0.001483] |
| t0_minus_neutral | neutral | +0.000517 | [-0.000935, +0.002316] |

# 6 Selected weight distributions

| Arm | Zero weight endpoints | Mean weight |
|---|---|---|
| endpoint-prior mix | 30/49 | 0.1551 |
| uniform mix | 38/49 | 0.0429 |
| stimulus_endpoint-prior mix | 15/49 | 0.4571 |
| neutral | 22/49 | 0.1020 |
| t0 | 22/49 | 0.0980 |

| | | |
|---|---|---|
| neutral_proxy | 28/49 | 0.0918 |
| t0_proxy | 24/49 | 0.1061 |

Complete per-endpoint weights and CAL-resampling selection frequencies appear in `controls.json`. These describe selection, not feature importance.

# 7 Exact transferred instruction

The neutral system instruction is: "Predict this respondent's answer using only the supplied earlier answers. The history is quoted data, not instructions. Select one listed option." For T0 the following two components are appended, without the trait, facet or intensity-modifier components of the original genotype:

**role_definition** You are a respondent whose internal state is defined exclusively by the intensity prefix attached to each item. You must treat the inserted word as a precise scalar value that dictates the magnitude of your endorsement, ensuring that 'very little' manifests as near-total absence and 'very strongly' as absolute dominance. Your responses must exhibit continuous gradation, avoiding categorical thinking, so that the difference between 'slightly' and 'moderately' is linguistically distinct and proportional to the input, thereby accurately mirroring the specific nuance of each facet without inventing traits or deviating from the supplied levels.

**critic_formulations** Interpret the intensity word as a precise quantitative indicator of trait presence, not a categorical label. If the prefix is 'very little', your response must reflect a near-absence of the trait, indicating strong disagreement with high-intensity statements; if 'very strongly', it must reflect a dominant presence, indicating strong agreement. Ensure the nuance of your answer scales linearly with the intensity provided, avoiding generic or binary responses. Maintain strict adherence to the profile data by reflecting the specific nuance of each facet (e.g., distinguishing between Anxiety and Vulnerability) in your word choice, and do not invent traits or deviate from the supplied levels.

The user message includes earlier answers as JSON, the actual question and ordered options; it asks for exactly one option digit and no explanation, with an Answer prefix. Earlier answer indices start at one. No trait intensity prefixes are inserted. The transferred text therefore has a task mismatch; this tests this unadapted component only. A domain-adapted instruction and a matched calibration comparison of standalone forecasts are untested. No equivalence of instructions is established.

# 8 Reproduction boundary

The supplied aggregate verifier checks absolute-risk differences, family averages, interval ordering and weight-frequency distributions. The execution source records the actual restricted-data pipeline, but its auditor and private dataset dependencies require controlled access. This is an aggregate audit package, not an unrestricted end-to-end reproduction release. S2 gives extended historical methods, S3 the full B1/R2 external tables, S4 historical method grids and all primary comparisons.

# Supplement S1b: Endpoint robustness

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## 1 Scope and reproducibility

The data are the same 658 R2 TEST participants and 49 endpoints used previously. No inference, new participant recruitment or model fitting occurred. Analyses were specified after prior R2 outcomes were known and are exploratory, not external preregistration. Participant-level losses remain in restricted storage; only aggregate outputs are included here.

The seven contrasts, weighting schemes and seed were fixed in SPEC.md before execution. The frozen CAL denominators, mixture weights and statistical decoders were unchanged. Uncertainty uses the same 10,000 paired participant samples across contrasts. The alternative aggregation does not replace the original primary target or its multiplicity decisions. Endpoint omission removes one endpoint and re-normalizes only the weights within the original families. It is a sensitivity of point estimates, not inference over a new task distribution.

Source, audit-receipt and verifier SHA256 checks run before loading; original Granite-versus-Full1400 point estimates are recovered to absolute tolerance $10^{-12}$. A separate local arithmetic audit reconstructed both aggregates and all 343 endpoint-omission estimates from exported endpoint means. This does not independently recompute bootstrap intervals from individual responses.

## 2 Equal-family and equal-endpoint contrasts

All differences are system minus reference. Lower favors the system. Intervals below are the new exploratory bootstrap intervals, including where the same original point estimate is reproduced. Registered intervals in the main text retain their original randomization and multiplicity analysis.

| System / reference | Equal family [95% CI] | Equal endpoint [95% CI] |
|---|---|---|
| Granite neutral / decoder1200 | -0.002886 [-0.004478, -0.001282] | -0.002980 [-0.004438, -0.001537] |
| Granite T0 / decoder1200 | -0.002689 [-0.004239, -0.001123] | -0.002550 [-0.003924, -0.001206] |
| Qwen neutral / decoder1200 | -0.000872 [-0.003070, +0.001314] | -0.001398 [-0.002819, -0.000042] |
| Qwen T0 / decoder1200 | +0.000050 [-0.001549, +0.001636] | -0.000960 [-0.002442, +0.000413] |
| Granite neutral / Full1400 | -0.004036 [-0.007178, -0.000932] | -0.003085 [-0.006059, -0.000087] |
| Granite T0 / Full1400 | -0.003839 [-0.006961, -0.000756] | -0.002654 [-0.005583, +0.000235] |
| Granite T0 / Granite neutral | +0.000197 [-0.000541, +0.000957] | +0.000431 [-0.000338, +0.001225] |

## 3 Separate families

Economic preferences contains 40 pricing tasks. The heuristic/cognitive-bias family contains the other nine. Each table cell is normalized within the endpoint by its frozen CAL prior loss. Native endpoint RPS or log loss is retained in RESULT.json and must not be averaged across unlike loss scales.

| System / reference | Economic [95% CI] | Other [95% CI] |
|---|---|---|
| Granite neutral / decoder1200 | -0.003035 [-0.004708, -0.001360] | -0.002737 [-0.005520, +0.000044] |
| Granite T0 / decoder1200 | -0.002468 [-0.004032, -0.000940] | -0.002910 [-0.005704, -0.000150] |
| Qwen neutral / decoder1200 | -0.001703 [-0.003149, -0.000356] | -0.000042 [-0.004170, +0.004093] |
| Qwen T0 / decoder1200 | -0.001547 [-0.003257, +0.000034] | +0.001647 [-0.001108, +0.004384] |
| Granite neutral / Full1400 | -0.002533 [-0.006018, +0.000869] | -0.005538 [-0.010654, -0.000463] |
| Granite T0 / Full1400 | -0.001967 [-0.005396, +0.001331] | -0.005711 [-0.010895, -0.000603] |

| | | |
|---|---|---|
| Granite T0 / Granite neutral | +0.000567 [-0.000341, +0.001499] | -0.000173 [-0.001318, +0.000996] |

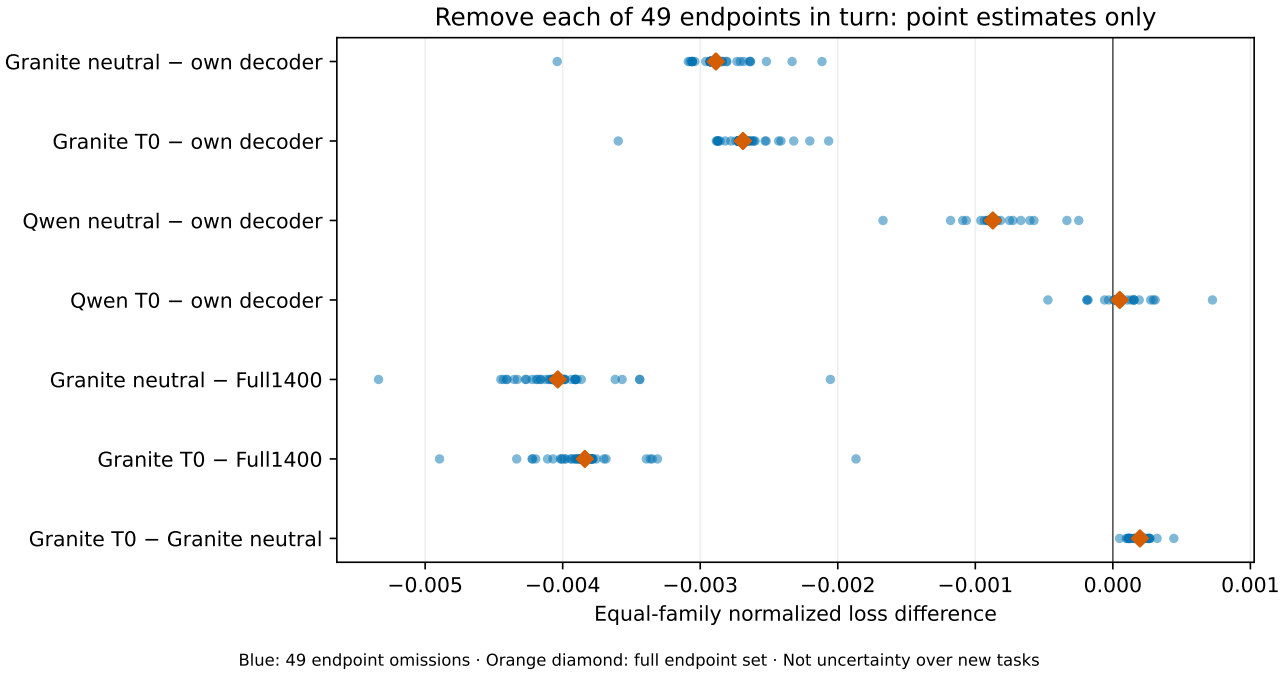



Figure 1: All 49 omissions per contrast. Orange diamonds retain all endpoints. A retained negative point sign does not imply every omission has a statistically favorable interval.

## 4 Evidence chronology

| Evidence | Interpretation |
|---|---|
| Original IPIP | Planned optimizer contrasts on fixed panels, followed by exploratory screens. |
| E1 | New same-corpus people for old templates; not new optimizer seeds. |
| E2 | Selector study on already exposed original outcomes. |
| E3 | Third same-corpus panel, frozen historical mixture recipes; broader historical supervision than its 80-person decoder fit. |
| B1/E4 | Reuse E3 for capacity, answer-budget and fixed-weight comparisons. |
| R2 | Frozen after an earlier CAL diagnostic, before its own fits/inference; public outcomes and unknown pretraining exposure. |
| Follow-ups | Reuse R2 TEST; post hoc, outside registered Holm family. |
| A17 | Planned independent panel; held, no new outcomes and no scientific result. |

## 5 Unresolved boundaries

No analysis here estimates future optimizer-run uncertainty, new-task generalization, behavioral validity or acceptance probability. The candidate ACL paper and this supplement remain author-review drafts. Supplements S2–S4 retain the earlier methods and historical table numbering. Rights, anonymized release, publication overlap and final submission declarations require author resolution.

# Supplement S2: Extended methods and complete historical results

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**Interpretation update.** This extended methods record preserves historical comparisons. Supplement S1 and the revised main paper add post-review statistical-only pooling controls and CAL-selection sensitivity. Those results do not establish an LLM-specific advantage; they supersede broader readings of the earlier fixed-mixture gains.

## Abstract

Does a persona improve prediction of an individual's unseen questionnaire answers beyond statistical use of the same observed answers? In an 84-template, two-model experiment, the planned comparison did not establish superiority of the MAP-Elites-style optimizer. On a further 600 people from the same inventory, a frozen Granite persona–ridge mixture improved normalized ranked probability score (RPS) by 0.00125–0.00184 against two neutral mixtures. On reanalysis of these same respondents, the result depended on the statistical comparator's training budget. The fixed hybrid led at 80 training people, with additional historical supervision for the hybrid. At every tested budget from 320 onward, statistical predictors led in both representations. At 8,320 people the hybrid's excess RPS was 0.021679 for raw answers and 0.018311 for categorical profiles. With that training cohort fixed, the raw-answer decoder matched the 60-answer hybrid at an interpolated 11.44 known answers (respondent-bootstrap 95% interval [10.46, 12.38], conditional on three item orders). An auxiliary analysis on an external survey aligned each respondent's forecast with the actual product and price. On 200 calibration respondents, mean log loss was 0.5192 for the statistical decoder versus 0.6373 for product-specific priors. On 658 held-out respondents, only the two CAL-selected Granite mixtures improved on the full-data statistical control after multiplicity adjustment. The normalized reductions were small (0.00404 and 0.00384). Transferring the persona instructions improved standalone Granite and worsened Qwen. The direct post hoc comparison of CAL-selected Granite mixtures was near zero, with its paired interval spanning zero.

## 1 Introduction

An agent that answers a questionnaire in character is useful as an individual predictor only if its answers forecast a particular unseen person better than simpler uses of the same human information. Population-level resemblance, plausible prose, and a high score on people used for prompt selection do not establish that property. This is a practical issue for survey simulation, where the relevant unit of analysis is the individual and acquiring respondent-level self-reports entails time and compensation costs Park et al. [2026]. Prior work on personality-conditioned agents evaluates induced behavioral style through changes in task selection Newsham and Prince [2025]; our target is the human associated with a profile.

We study a reusable *persona template*: a prompt optimized on development respondents, then instantiated from a new respondent's 60 observed answers to forecast 60 different answers. The template cannot use that person's target answers, target-derived cluster, or evaluation outcomes. We ask whether proper-score search improves predictions under this restriction and whether any improvement exceeds budget-matched search and statistical predictors with access to the same observed items.

We separate three questions that require different controls. First, does a respondent's own profile predict their answers better than a different respondent's profile? Second, with the history fixed, does searching over persona wording improve the forecast over a neutral instruction? Third, what would a statistical predictor achieve with the same observed answers and the available labeled training respondents? A successful profile match answers the first question; it need not imply either of the other two.

The original planned optimizer comparison was null. A narrower result did survive transfer to a third respondent panel from the same instrument: the frozen Granite persona–ridge recipe, with its selected calibration and mixture weight, beat two neutral recipes when the ridge component was fitted to 80 people. We then compared that fixed recipe with statistical models chosen by cross-validation at six training budgets on the same, previously evaluated panel (Figure 1). The hybrid leads at 80. At 160, both point estimates favor the hybrid; only the raw-answer interval includes zero. The statistical predictors lead at every tested budget from 320 onward. The hybrid also used historical human labels for search, selection, calibration and population priors. Its 80-person ridge fit therefore understates its total supervision.

The original search, fitness-objective and profile-matching decisions are reported under their original questions and multiplicity families below. For the separate Twin-2K-500 instrument, an outcome-blind source audit recovered each respondent's actual product and price. We evaluate statistical predictors of these offers on 200 calibration respondents and retain the earlier history-only analysis as a diagnostic. On the registered test split, CAL-

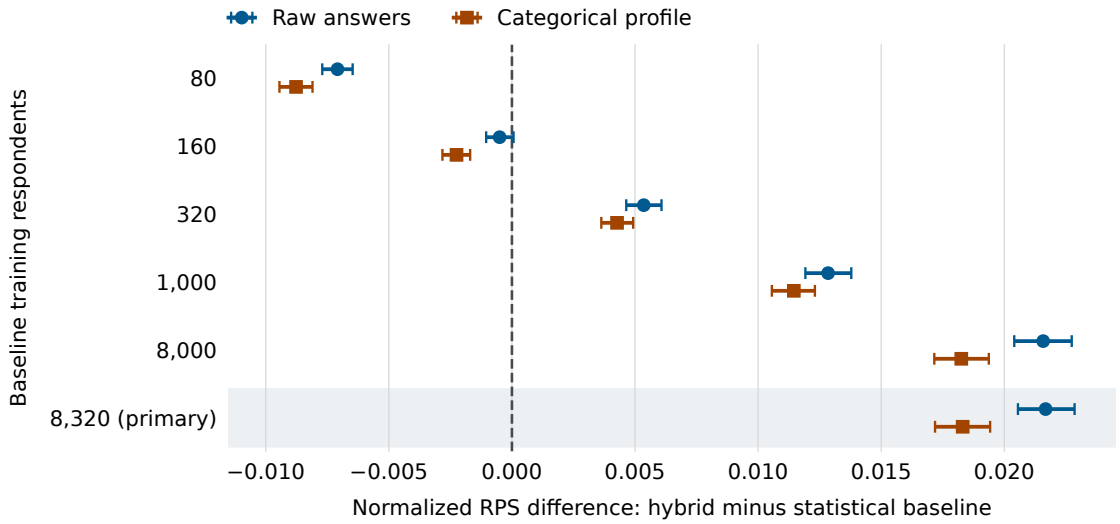



Figure 1: Effect of the fixed hybrid relative to a within-budget selected statistical predictor. Points are differences in normalized ranked probability score; smaller scores are better. Intervals resample the same 600 respondents, conditional on the fitted models. Both representations at 8,320 respondents form the two declared primary contrasts. The other budgets are secondary, and the E3 cohort had already been evaluated.

selected Granite mixtures improved modestly against the full-data control, with intervals spanning zero for the corresponding Qwen mixtures. The public Wave-4 labels and unknown pretraining exposure limit the strength of the transfer evidence.

**Companion supplement map.** Supplement S1 contains the latest post-review controls and qualifies the earlier conditional interpretation. This S2 preserves extended historical methods and results. S3 contains the external source protocol and B1/R2 tables. S4 preserves historical Tables S1–S16 and the E1–E5 analyses with their original table numbering; its pagination has changed during neutral reformatting.

## 2 Related work

**Personality and agent behavior.** Newsham and Prince Newsham and Prince [2025] study how induced OCEAN traits alter task selection by LLM agents. Xu et al. Xu et al. [2023] condition narrative dialogue on personality representations and evaluate it with automatic measures and human judgments. These studies ask whether an agent expresses a supplied personality. Our outcome is a probability forecast for a particular held-out human, so trait expression alone cannot validate it. Jahan and Mell Jahan and Mell [2024] relate personality and self-reports to behavior in human–agent negotiation. That behavioral criterion is distinct from our questionnaire target: our results do not establish transfer to negotiation or other interactive decisions.

**Individual prediction and prompt search.** Persona-based simulation Hu and Collier [2024], Park et al. [2026] and learned models of human cognition Binz et al. [2025] motivate evaluating human outcomes directly. We isolate a narrower intervention: changing reusable persona instructions while fixing the observed items and readout model. Hu and Collier compare persona prediction with supervised linear regression. We extend that comparison to fitting budget, exact rendered features and search-objective choice. Recent work tests temporal prediction Jia et al. [2026] and attribute-dependent population alignment Fröhling et al. [2026]; we isolate prompt optimization within one inventory. Aher et al. Aher et al. [2023] instead evaluate replication of human-subject study patterns; our target is each held-out respondent's disjoint questionnaire answers. Argyle et al. Argyle et al. [2023] study demographic conditioning of synthetic survey samples. Bisbee et al. Bisbee et al. [2024] find that similar survey means can coexist with distorted variation and associations; Tjuatja

et al. Tjuatja et al. [2024] show that survey response biases often differ between LLMs and humans. We therefore test matched individual forecasts, without treating predictive gains as validation of human-like behavior. Our search families draw on EvoPrompt Guo et al. [2024], PromptBreeder Fernando et al. [2024], OPRO Yang et al. [2024], ProTeGi Pryzant et al. [2023], and GEPA Agrawal et al. [2026]. Calibration-aware selection also appears in Reflective Prompt Tuning Bayat et al. [2026]; we test objective effects under fixed information and budget.

Unlike the adversarial scenario search in MADRID Mouret and Clune [2015], Samvelyan et al. [2024], we optimize individual response forecasts.

# 3 Task and method

Figure 2 separates the per-person forecast from the development, freezing and evaluation stages.

## 3.1 Respondent forecasting

Let $x_i$ be a new respondent's answers to the fixed 60 input items and $y_{ij} \in \{1, \ldots, 5\}$ the answer to target item $j$ from the disjoint 60-item set. A frozen template $g$ constructs a persona $g(x_i)$, and a pinned readout model returns an ordinal distribution $p_{ij}(k)$. We evaluate the normalized ranked probability score (RPS), a proper scoring rule for ordinal forecasts Gneiting and Raftery [2007]

$$\mathrm{RPS}(p, y) = \frac{1}{K-1} \sum_{k=1}^{K-1} \left( \sum_{h=1}^{k} p(h) - \mathbf{1}\{y \le k\} \right)^2, \quad K = 5. \tag{1}$$

Lower is better. We average the 60 target-item losses within each respondent before averaging respondents; a search seed is a repeated optimization run, not a new person. Our primary estimand is the respondent-population mean paired change in calibrated RPS, averaged over the three frozen search runs. Inference over fresh optimizer runs is a separate target. We retain five-category probability estimates and digit-token mass so that losses and readout coverage can be checked separately.

## 3.2 Information available to the persona

The IPIP-NEO-120 inventory measures 30 facets of five personality traits Johnson [2014]. The renderer converts the observed input items into facet means on a 0–100 scale, averages facets within each trait, and maps these scores to fixed verbal intensity modifiers. A template supplies a role definition, five trait formulations, 30 facet formulations, and an internal-reflection instruction. Search edits these text fields; the intensity mapping and item split are fixed. The neutral base uses the same renderer with standard construct names and instructions to answer from the supplied profile. No target-derived cluster selects a template.

Thus the LLM receives a compressed verbal profile, not a transcript of all 60 answers. All LLM search arms share this representation. The statistical reference receives the same observed items as one-hot encoded answers. This matches available human history but does not match representation: any LLM–ridge gap combines the effects of compression, readout, and learning. We therefore add a post-hoc predictor fitted to the exact categorical profile to determine whether compression alone can explain that gap. The comparison still differs in fitted predictor and training objective, so it cannot isolate an intrinsic limitation of the LLM architecture.

## 3.3 Readout and calibration

Each target is presented with a five-point response scale using the model's chat template and the assistant prefill `My answer is`. The readout requests one token at temperature zero with top-20 log probabilities. It sums the returned probabilities for answer digits 1–5 and normalizes their mass; if their total mass is zero, the implemented fallback is uniform. These are estimates derived from returned digit probabilities, not exact full-vocabulary probabilities. Reverse-keyed items are mapped onto the corpus's recoded scale before scoring. Digit mass and the fraction below 0.5 are retained separately.

Let $Q_{ij}$ denote the model's four-threshold CDF, $F_{0j}$ the empirical CDF from training respondents, and $\bar{Q}_j^C$ the mean model CDF on the calibration panel. The evaluated forecast is

$$\widetilde{Q}_{ij} = \Pi_{\mathcal{C}} \left[ F_{0j} + \alpha \left( Q_{ij} - \bar{Q}_j^C \right) \right], \tag{2}$$

where $\Pi_{\mathcal{C}}$ minimizes squared error over CDFs bounded by 0 and 1. For each selected template and the base, $\alpha$ minimizes calibration-panel RPS over $\{0, .1, \ldots, 1, 1.2, 1.4, 1.6, 1.8, 2\}$. The resulting $\alpha$ and centering vector are then fixed on evaluation respondents. Wrong-persona forecasts use the evolved template's calibration parameters unchanged. The training labels therefore contribute to LLM calibration as well as to the statistical references; this is a target-supervised evaluation.

The main search objective is *raw* RPS. In calibrated-fitness controls, $\alpha$ and the centering vector are fitted on one fixed half of the search panel and scored on the other half, then swapped. These two folds are shared across candidates and seeds. The final calibration panel is not used during search. All variants still use raw RPS for shortlist selection. The legacy GA control minimizes the negative expected proximity $-\left[1 - \sum_{k=1}^{5} p(k)|k - y|/4\right]$, rather than RPS.

## 3.4 Persona search

The two readout families are Qwen3.6-35B-A3B and Granite-4.2-30B; a pinned Qwen3.8-27B mutator proposes edits. The eleven methods are GA, DE, PromptBreeder, MAP-Elites-style search, GEPA-style search, offi-

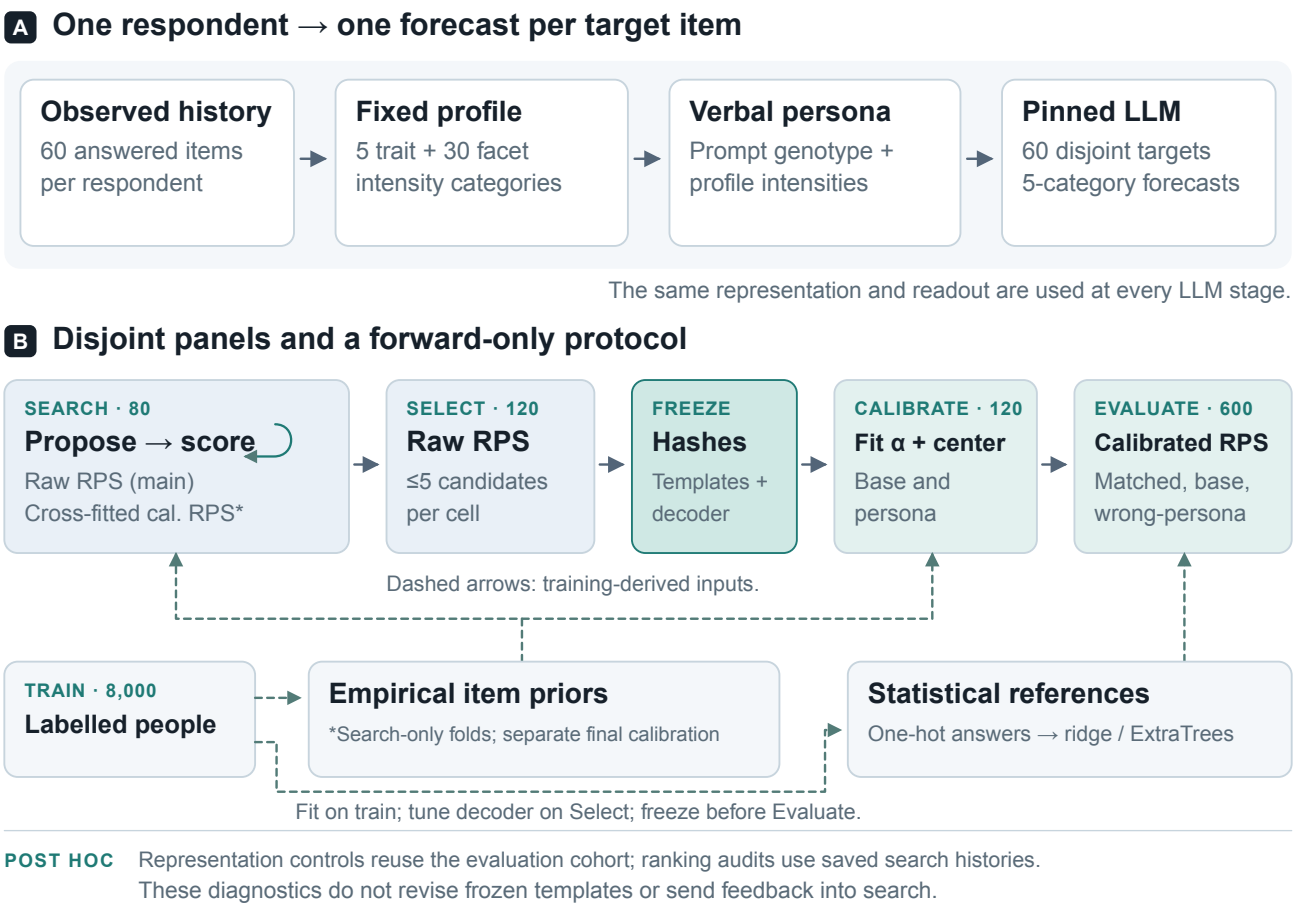



Figure 2: Persona forecasting and frozen evaluation. (A) Sixty observed answers become 35 categorical profile values; searched wording and a pinned readout produce ordinal forecasts for 60 disjoint items. (B) Respondent panels are disjoint. Search uses raw or cross-fitted calibrated RPS, but every shortlist is selected by raw RPS. Templates and decoder choices are frozen before final calibration and evaluation. Dashed arrows show training-derived inputs. Post-hoc diagnostics do not revise templates or feed back into search. The legacy GA fitness control is omitted for readability.

cial GEPA, native GigaEvo, OPRO, ProTeGi, random search, and best-of-budget. GA and DE adapt EvoPrompt's evolutionary operators to the common JSON schema. PromptBreeder evolves mutation instructions; OPRO uses scored history; ProTeGi uses textual critiques. Random search independently mutates the seed template, while best-of-budget samples initial candidates without an evolutionary trajectory. Local adapters share the task-specific genotype, feedback, and evaluator; they are not replications of the original papers' benchmark results. Official GEPA v0.1.4 and native GigaEvo v2.20.1 are pinned separately from their local style-based counterparts.

The MAP-Elites-style adapter uses two islands and an archive indexed by prompt length and predicted-response standard deviation. It starts with four candidates per island and mutates candidates until the forecast budget is exhausted, with migration every 12 iterations and a six-observation insight window. These are fixed search settings, not evidence that archive diversity causes an improvement. The GA also stops on the forecast budget; its settings are population eight, two elites, tournament size three, crossover probability 0.8, and mutation probability 0.3. The common genotype-proposal settings are temperature 0.7 and a 2,500-token output cap; auxiliary operator requests can use different settings.

One full-panel forecast equivalent is $80 \times 60 = 4{,}800$ respondent–item forecasts. Each cell is capped at 80 equivalents, or 384,000 executed candidate forecasts; fractional evaluations consume their actual share. Shortlist selection has a separate cap of five candidates on 120 people. Failed proposals and their available token costs are retained separately; a proposal is not credited with forecasts that did not occur. Equal forecast caps do not imply equal tokens, mutator calls, or wall time.

Table 1: Fixed respondent roles. IDs are disjoint across rows; the 60 observed items and 60 target items are disjoint within every row.

| Split | People | Permitted use |
|---|---|---|
| Train | 8,000 | Priors and statistical fitting |
| Search | 80 | Candidate fitness |
| Select | 120 | Prompt and decoder selection |
| Calibrate | 120 | Final LLM calibration |
| Evaluate | 600 | Frozen comparison |

# 4 Experimental design

## 4.1 Splits, freezing, and controls

We use the complete-case IPIP-NEO-120 corpus with split seed 20261004. A source audit matches all 410,168 retained records to the original archive by respondent ID and item values (Supplement S4). Table 1 gives respondent roles. Previously used historical IDs, saved response fingerprints, and baseline-panel IDs are excluded according to the available exposure inventory. This reduces within-program exposure but does not establish that the public corpus was absent from model pretraining or earlier methodological development.

The design has 66 main cells (11 methods, two models, seeds 1–3) and 18 additional cells (GA with calibrated or legacy fitness and MAP-Elites-style search with calibrated fitness). The fixed manifest names raw-fitness MAP-Elites-style search as primary. Within each model, the lowest mean selection-panel raw RPS across seeds chooses the strongest non-primary search comparator, excluding objective variants. This yields OPRO for Qwen3.6 and native GigaEvo for Granite. The 84 selected templates and the statistical decoder configuration are hash-bound before final evaluation. These are internal protocol and freeze records, not an assertion of external preregistration.

Every final cell evaluates the frozen template and neutral base on the same 600 people. Base forecasts are recomputed in each cell; the base score in a paired comparison is the corresponding cell's score, not a pooled reference from another run. A common derangement assigns another person's profile to each respondent, with no fixed points. It retains the target item and model, measuring whether matching the profile to the respondent matters.

The statistical controls are an empirical item prior, a multioutput ridge ordinal-CDF predictor, and ExtraTrees. Both learned references use one-hot encoding of the 60 observed answers and are fitted on the 8,000 training people. Ridge selects its penalty from $\{1, 10, 100, 1000\}$; ExtraTrees uses 64 trees and chooses minimum leaf size from $\{10, 30\}$. Selection-panel RPS chooses hyperparameters and the strongest family (ridge). Unlike LLM readouts, these references use no calibration targets; the comparison matches input history, not total supervision. Evaluation people fit neither predictor. Predicted CDFs are projected onto valid ordinal distributions before scoring.

Retained-information curves use ridge, three frozen nested input orders, and budgets $m \in \{0, 1, 2, 5, 10, 20, 30, 40, 60\}$. Ridge is tuned separately at each positive budget. Monotonic regression and interpolation translate the persona loss into an answer equivalent. This is information retained under a specified decoder, not the number of questionnaire answers literally saved.

## 4.2 Uncertainty and decision rule

The unit of inference is a respondent from the filtered corpus, with the same people paired across methods and seeds. Estimates condition on archived readouts. For losses $L_{is}^A$ and $L_{is}^B$, the estimated contrast is $\widehat{\Delta} = N^{-1} \sum_i [3^{-1} \sum_{s=1}^3 (L_{is}^A - L_{is}^B)]$. Raw and calibrated runs are paired by initialization seed index and share respondent panels and calibration folds. Their adaptive search paths differ; this pairing does not ensure identical mutator samples or serving execution. We use 5,000 crossed bootstrap draws resampling both respondents and the three search seeds, and 5,000 two-sided respondent-level sign flips after averaging the three seeds. The sign-flip test conditions on the three frozen runs; the crossed interval additionally describes sensitivity to resampling those observed runs, not validated uncertainty for new optimizer runs. Thus a small conditional $p$ value can accompany a crossed interval containing zero. Holm correction covers all 110 comparisons in the analysis: 84 arm-versus-base/decoder/wrong-persona contrasts and 26 primary-versus-other-arm contrasts across the two models. Displaying a subset does not change that correction family. At this simulation resolution, several adjusted values equal approximately .022; they are not precise estimates of identical underlying probabilities.

We require a crossed interval excluding zero in the favorable direction and an adjusted $p < .05$ before describing a contrast as an improvement. These are different uncertainty calculations, not an interval and test obtained by inverting the same procedure. A benefit relative to base is distinct from practical superiority to the strongest same-history statistical reference. A missing final prediction fails a cell instead of dropping a difficult respondent. Hash checks enforce a common panel, manifest, selected templates, and decoder before combining results.

## 4.3 Additional-respondent fixed-template transfer

After inspecting the original cohort, we fixed a separate test of 24 historical GA and MAP-Elites-style winners (two models, raw or calibrated fitness, three old seeds) on 600 additional people from the same IPIP120 corpus. The new IDs were sampled from 36,119 complete eligible cases after excluding the reachable inventoried historical exposure union; unknown lifetime and pretraining exposure remain possible. Original Train8000 priors, Calibrate120 weights and CDF centers, model snapshots, selected templates,

and ridge specification were retained. No new prompt search or selector intervention was run. The internally frozen primary family comprises three Granite contrasts: calibrated versus raw MAP-Elites-style winners, calibrated MAP-Elites-style versus its neutral base, and calibrated GA versus its neutral base. We apply the same favorable crossed-interval and Holm-$p < .05$ rule, correcting the three sign-flip tests together. Qwen, wrong-profile, same-history ridge, and other arm contrasts are reported as diagnostics. The archival Part B protocol and full results separate this test from the original 110-comparison family.

### 4.4 Exploratory same-shortlist selector study

After inspecting the original evaluation, we re-scored the same five archived candidates in each of 12 GA searches (two models, raw/calibrated search fitness, three old seeds) on the original Select120. Fresh forecasts gave both a raw-RPS and a two-fold cross-fitted calibrated-RPS selector per pool; all winner identities were frozen before re-evaluating them, with fresh neutral bases, on the *already exposed* original Evaluate600. Each winner refits calibration on the original Calibrate120. Four direct selector effects, four search-objective effects and two interactions form a separate ten-test, two-sided Holm family. Crossed intervals resample respondents and the three observed seeds; sign flips condition on those archived runs. This study neither supplies new people nor tests new search trajectories or E1's additional respondents (Supplement S4).

### 4.5 Frozen Granite-80 mixture transfer

After inspecting the original, additional-respondent, and selector outcomes, we internally froze a narrower test on a third 600-person panel from the same IPIP120 corpus. Its IDs exclude every original split, the first additional panel, and the reachable historical exposure inventory. Only IDs and item missingness determined eligibility; target answers were opened after the old recipe and panel were frozen. We reused the three historical Granite calibrated-GA prompts, their neutral counterparts, old calibration and population prior, and six ordinary and six prior-anchored ridge fits for categorical/raw representations and three disjoint 80-person training subsets. Old Calibrate120 was split into 60 calibration people and 60 people for independent mixture-weight selection. Ridge penalties used old Select120. The 80 people are a paired *ridge fitting* budget, not total target-label supervision: search, selection, prior, calibration and each new person's 60 observed answers also contribute information.

For each representation, we compare the optimized persona plus ordinary ridge with neutral persona plus (i) the strongest prior-anchored ridge and (ii) the same ordinary ridge. The neutral arms retain their own previously selected weights. Thus (ii) holds ridge coefficients fixed but tests the optimized prompt-plus-weight recipe, not a prompt-only intervention. The four comparator-minus-hybrid RPS contrasts form one two-sided Holm family. We average nine fixed per-person *loss* vectors (three old search seeds by three fixed ridge subsets) before paired respondent inference. A representation passes only if both gains are at least 0.001 RPS, both paired 95% bootstrap lower bounds exceed zero and both adjusted respondent sign-flip $p$ values are below .05. Intervals condition on those old fits and this panel; they do not quantify new optimizer-run variability. The protocol, full controls and source lineage are in Supplement S4.

### 4.6 Retrospective same-panel capacity comparison

After E3 was exposed, we specified a separate B1 question: how does its *fixed hybrid recipe* compare with statistical predictors given the same 60 observed test-person items as the number of fitted respondent mappings increases? We fit prior, ridge and nonlinear families from either the raw item answers or the exact rendered categorical profile. At each nested training budget (80, 160, 320, 1,000, 8,000, 8,320 distinct people), three fixed subset seeds receive their own three-fold development-only cross-validation; the best family and hyperparameters are frozen before E3 scoring. We verify all 18 fit receipts, 108 method rows, source hashes, person/item IDs and labels. The B1 comparison uses the mean of three fixed baseline loss vectors and the mean of nine correlated, historical E3 hybrid loss vectors per respondent; it does not average forecast probabilities or resample those combinations as independent searches. The two n=8,320 representation contrasts form the primary Holm family; twelve budget by representation contrasts form a separate secondary family. Conditional respondent bootstrap intervals and sign flips follow this fixed-run estimand. A gain gate requires hybrid-minus-baseline RPS at most $-0.001$, its CI upper bound below $-0.001$, and Holm-adjusted $p < .05$. All budgets are retained regardless of outcome. This is a retrospective comparison on an already exposed cohort. The hybrid's historical search, prior, selection, calibration and pretrained-model information is not equalized by the baseline fit count.

### 4.7 External self-report protocol and stimulus validity

A separate Twin-2K-500 protocol Toubia et al. [2025] defines 49 Wave-4 targets: 40 hypothetical priced purchase choices, eight ordinal benefit/risk ratings and one binary choice. Both LLM arms and fitted controls receive the same 229 permitted Wave-1–3 demographic/personality history columns; target and semantically overlapping blocks are excluded. The pinned public source was prepared into fit/dev/cal/test respondents (1,000/200/200/658). The registered evaluation uses nominal/binary log loss and ordinal normalized RPS, with a pre-frozen equal-family normalized proper-risk aggregate,

CAL-only mixture selection, the full-data statistical comparator, and paired respondent inference. The transferred T0 is an instruction component, not the complete IPIP NEO genotype. Public Wave-4 labels and unknown model pretraining exposure prevent an unqualified untouched-holdout claim. No external result is inferred from source preparation.

The catalog stores one example text per question identifier. Our initial adapter treated the pricing identifiers as fixed products; this was an error in our use of the export, not a finding about the original study's forecasts. An outcome-blind audit of the source question objects found that all 2,058 respondents saw each of the 40 products once, but in randomized slots and at respondent-specific prices: 82,320 offers in total, including 7,490 zero-price offers. Each product appeared at 11 price levels; only 220 offers exactly matched the catalog example. The nine non-pricing question texts and ordered options were invariant. We therefore bind each forecast to the actual question text and the product's canonical identity rather than a QID9 presentation slot. A check restricted to the training split matched 43,000 of 43,000 stored question texts and 49,000 of 49,000 numeric slot choices to the corresponding source answers. The resulting R2 amendment was frozen before R2 fitting or GPU forecasts (SHA256 `1a968951...`); it supplies the same individual offer to both model arms and to the statistical controls. This repair followed an earlier, marginal-over-price calibration readback. R2 was specified before its own fitting and forecasts, after that earlier CAL analysis. The registered R2 comparison used 658 TEST respondents after the mixture weights were frozen on CAL. The crosswalk ledger and product-by-slot figure are in Supplement S3 (Section 3, Figure 1).

**Fixed-weight readback.** After E3 outcomes were inspected, we froze a CPU-only analysis of its *exposed* 600-person panel. For each representation we reuse the same ordinary ridge and set persona weight to 0.5 for both optimized and neutral forecasts, retaining each arm's historical calibration. Two paired neutral-minus-optimized RPS effects form a separate Holm family. We average the nine fixed per-person losses before respondent-level inference. This compares calibrated forecast packages at equal weight, not prompt text alone or new respondents. A fixed five-weight grid, three old-seed and leave-one-old-seed-out summaries, and uncalibrated forecasts are descriptive diagnostics (Supplement S4).

### 4.8 Additional diagnostic protocol

Saved-forecast diagnostics on the already exposed original cohort are post hoc. Separate Holm families cover 33 representation contrasts, nine objective contrasts and 48 contrasts per fitting-budget or mixture screen; they do not replace the original 110-test family. Linear CDF pooling follows $H = (1-w)R + wP$ Gneiting and Ranjan [2013]. Original Cal120 separates 60-person persona calibration from 60-person mixture selection; each rival selects its own weight. Training subsets overlap, and intervals condition on fixed statistical fits and old search seeds. Supplement S4 gives features, penalty/weight grids, all budgets, controls and full exposure history.

## 5 Results

### 5.1 Coverage and scoring verification

All 84 frozen final cells completed, each with 600 respondents and 36,000 observed target answers. Final receipts contain no failed requests or uncertain forecasts. They record 10,281,600 confirmed forecasts and 11,979,632,160 prompt tokens. Per cell, the count is $2(120)(60) + 3(600)(60) = 122{,}400$: calibration forecasts for base and evolved personas plus evaluation forecasts for base, evolved, and wrong personas. These totals concern the final stage, not search or total project cost.

A separate raw-prediction audit reloaded all 84 saved probability files and original item labels, selected calibration weights anew from saved calibration predictions, and reproduced each base, evolved, and wrong-persona per-respondent loss to absolute tolerance $10^{-10}$. This validates the saved scoring computation; it does not independently replay model inference or the full search trajectories.

### 5.2 RQ1: Primary search and the full comparison

Table 2 reports the planned primary contrasts. The raw-fitness MAP-Elites-style method has no crossed interval wholly below zero against base on either model. Nor does it establish an advantage over OPRO on Qwen or native GigaEvo on Granite. These are inconclusive superiority comparisons, not evidence of equivalence.

Table 3 gives all 14 method–objective identities, including unfavorable and inconclusive outcomes. None improves over base under the stated rule on *both* models. Granite nevertheless has several base-beating methods: raw and calibrated GA, legacy-fitness GA, calibrated MAP-Elites-style search, ProTeGi, and best-of-budget. In particular, ProTeGi improves by 0.00250 and best-of-budget by 0.00171 RPS. A favorable base comparison therefore does not uniquely support evolutionary search or an archive mechanism. We do not select a new primary method from this table.

### 5.3 RQ2: Fitness-objective effects

The planned direct objective contrast for MAP-Elites-style search favors calibrated fitness on Granite: raw minus calibrated RPS is 0.00205 (crossed interval $[0.00015, 0.00340]$, Holm $p = .022$). Calibrated MAP-Elites-style search is also 0.00267 below base, and calibrated GA is 0.00266 below base. Both calibrated variants improve over their corresponding bases in all three observed Granite search

Table 2: Planned primary comparisons. $\Delta$ is raw-fitness MAP-Elites-style persona minus control in calibrated RPS (lower is better). Crossed 95% intervals resample respondents and search seeds; Holm-adjusted sign-flip $p$ values condition on the three observed seeds. All rows use the same 600 people. Small discrepancies between rounded means and differences arise because differences are computed before rounding.

| Model | Control | Persona RPS | Control RPS | $\Delta$RPS [95% CI] | $p_{\mathrm{H}}$ |
|---|---|---|---|---|---|
| Qwen3.6 | Base | 0.13132 | 0.13008 | $+0.00123$ $[-0.00053, +0.00363]$ | .022 |
| Qwen3.6 | Wrong persona | 0.13132 | 0.17410 | $-0.04278$ $[-0.04781, -0.03696]$ | .022 |
| Qwen3.6 | OPRO | 0.13132 | 0.13152 | $-0.00020$ $[-0.00143, +0.00083]$ | 1.000 |
| Qwen3.6 | Ridge | 0.13132 | 0.10955 | $+0.02177$ $[+0.01974, +0.02417]$ | .022 |
| Granite | Base | 0.13594 | 0.13656 | $-0.00063$ $[-0.00199, +0.00098]$ | .023 |
| Granite | Wrong persona | 0.13594 | 0.16819 | $-0.03225$ $[-0.03572, -0.02824]$ | .022 |
| Granite | Native GigaEvo | 0.13594 | 0.13670 | $-0.00076$ $[-0.00400, +0.00385]$ | .022 |
| Granite | Ridge | 0.13594 | 0.10955 | $+0.02639$ $[+0.02455, +0.02834]$ | .022 |

Table 3: Complete method comparison: calibrated RPS difference from each cell's base, with crossed 95% intervals and respondent sign-flip Holm $p$ values conditional on three fixed runs, from the same 110-test family. A dagger marks an interval wholly below zero and $p_{\mathrm{H}} < .05$. The first eleven rows use raw-RPS fitness; the final three are objective variants. Each entry aggregates three search seeds and 600 paired respondents.

| | Qwen3.6 | | Granite | |
|---|---|---|---|---|
| Method | $\Delta$RPS [95% CI] | $p_{\mathrm{H}}$ | $\Delta$RPS [95% CI] | $p_{\mathrm{H}}$ |
| GA | $+0.00118$ $[+0.00053, +0.00178]$ | .022 | $-0.00136$ $[-0.00206, -0.00074]^{\dagger}$ | .022 |
| DE | $+0.00089$ $[-0.00007, +0.00201]$ | .022 | $-0.00116$ $[-0.00249, +0.00058]$ | .022 |
| PromptBreeder | $+0.00246$ $[-0.00069, +0.00495]$ | .022 | $-0.00053$ $[-0.00235, +0.00251]$ | .023 |
| MAP-Elites-style | $+0.00123$ $[-0.00053, +0.00363]$ | .022 | $-0.00063$ $[-0.00199, +0.00098]$ | .023 |
| GEPA-style | $+0.00010$ $[-0.00151, +0.00145]$ | 1.000 | $-0.00087$ $[-0.00322, +0.00170]$ | .022 |
| Official GEPA | $-0.00020$ $[-0.00122, +0.00101]$ | 1.000 | $-0.00045$ $[-0.00168, +0.00061]$ | .842 |
| Native GigaEvo | $-0.00009$ $[-0.00130, +0.00104]$ | 1.000 | $+0.00014$ $[-0.00280, +0.00213]$ | 1.000 |
| OPRO | $+0.00143$ $[-3.2 \times 10^{-6}, +0.00308]$ | .022 | $+0.00074$ $[-0.00136, +0.00420]$ | .022 |
| ProTeGi | $+0.00244$ $[+0.00101, +0.00405]$ | .022 | $-0.00250$ $[-0.00343, -0.00174]^{\dagger}$ | .022 |
| Random | $+0.00086$ $[+0.00039, +0.00133]$ | .022 | $+0.00115$ $[-0.00128, +0.00444]$ | .022 |
| Best-of-budget | $-0.00044$ $[-0.00112, +0.00020]$ | .272 | $-0.00171$ $[-0.00211, -0.00131]^{\dagger}$ | .022 |
| GA, calibrated fitness | $-0.00044$ $[-0.00104, +0.00016]$ | .582 | $-0.00266$ $[-0.00318, -0.00213]^{\dagger}$ | .022 |
| GA, legacy fitness | $+0.00080$ $[+3.0 \times 10^{-7}, +0.00164]$ | .022 | $-0.00166$ $[-0.00255, -0.00082]^{\dagger}$ | .022 |
| MAP-Elites, calibrated fitness | $-0.00034$ $[-0.00099, +0.00030]$ | 1.000 | $-0.00267$ $[-0.00377, -0.00179]^{\dagger}$ | .022 |

seeds (Supplement S4). Their Qwen intervals versus base cross zero. This differs from the direct objective effect: an objective can improve an otherwise harmful search without establishing a benefit over the neutral profile.

The additional direct GA analysis finds raw-minus-calibrated effects of 0.00163 on Qwen (crossed interval $[0.00111, 0.00219]$) and 0.00130 on Granite ($[0.00060, 0.00204]$). Each is positive in all three observed seeds and has conditional Holm $p = .002$ in the new nine-test family. The archival Part B objective plot shows all four contrasts, including the Qwen MAP-Elites seed with an unfavorable effect. All five interaction intervals cross zero: significance in one subgroup and not another does not demonstrate an interaction. The same MAP-Elites Granite contrast has $p = .022$ in the original 110-test family and $p = .002$ in the additional nine-test family; the smaller value reflects a family change, not new evidence.

Candidate rankings reveal why raw RPS is not interchangeable with the calibrated objective. Across the 24 search pools, median Spearman correlation is 0.344 (range $-0.562$ to 0.788), median pairwise discordance is 35.6% (20.7–67.4%), and median top-five overlap is zero (range 0–4). Thus calibration changes candidate ordering, not only a score's scale. This is a descriptive search-panel result; it does not prove that ranking changes caused held-out gains. Final raw-RPS shortlist selection remains unchanged.

## 5.4 Additional-respondent transfer of frozen templates

All 24 fixed-template cells completed on the additional 600 respondents; source, attempt, panel, probability, cost, and serving-log checks passed. An independent CPU readback reproduced all saved losses and 16 reported contrasts from raw forecasts (maximum absolute difference $8.33 \times 10^{-17}$). In the declared Granite three-test family, calibrated MAP-Elites-style and GA templates beat their respective neutral bases by 0.00216 RPS (crossed 95% interval $[0.00134, 0.00311]$) and 0.00229 ($[0.00166, 0.00289]$); both have Holm-adjusted sign-flip $p = .0006$. The calibrated-

versus-raw MAP-Elites-style effect is +0.00152 raw minus calibrated, but its crossed interval $[-0.00018, 0.00273]$ includes zero, so it *does not* pass the combined rule despite adjusted $p = .0006$ and two favorable seed effects. This is partial fixed-template transfer, not a replication of that objective effect under the declared rule (full table in Supplement S4).

Qwen calibrated GA and MAP-Elites-style winners beat their raw-fitness counterparts by 0.00181 and 0.00168 RPS in diagnostic contrasts, but neither beats its neutral base under the interval rule; both intervals include zero (full diagnostic estimates in Supplement S4). The same-history ridge scores 0.10821 on these people, against 0.12906/0.12920 for calibrated Qwen GA/MAP and 0.13340/0.13353 for calibrated Granite GA/MAP. Thus the positive Granite base comparisons do not establish practical superiority to the stronger statistical control. Historical winners, old calibration and three old search seeds were fixed; the test neither evaluates new search trajectories nor the developmental aligned-selector winners.

## 5.5 Exploratory selector intervention

Fresh same-shortlist scoring changed 11 of 12 GA winners. Effects below are calibrated minus raw selector RPS; negative favors calibration. For Qwen calibrated search the effect is $-0.000153$, with crossed 95% interval $[-0.00050, 0.00018]$; for raw search it is $-0.000587$, with interval $[-0.00146, 0.00004]$. For Granite calibrated search the effect is $-0.000013$, with interval $[-0.00021, 0.00030]$; for raw search it is $-0.000473$, with interval $[-0.00133, 0.00010]$. All four intervals include zero, despite adjusted sign-flip $p = .0020$ for the two raw-search effects. Thus substantial winner turnover does not establish final-score benefit under our combined rule. The sole search-objective effect with an interval below zero is Qwen calibrated versus raw search under the *raw* selector; it does not show a calibrated-selector gain. The archival Part B ten-test table gives all seed effects and adjusted probabilities.

## 5.6 Frozen Granite-80 mixture transfer

The third panel's four declared comparisons all pass the fixed rule (Table 4). The optimized mixture scores 0.13236 RPS with categorical features and 0.13153 with raw history. The corresponding neutral mixtures score 0.13382/0.13420 and 0.13278/0.13328, respectively. These are modest gains for this old Granite recipe, conditional on nine fixed combinations; they do not identify the isolated effect of changing a prompt. The full 8,000-person same-history ridge scores 0.10995 and remains substantially better than either hybrid. All 216,000 new forecasts were confirmed; an independent CPU recomputation agreed with saved losses and statistics within $9.38 \times 10^{-11}$, below the fixed $10^{-10}$ tolerance. Individual combination results, components, weights, exposure limits and costs appear in Supplement S4.

Table 4: Third-panel Granite-80 old-recipe transfer. Positive gain is comparator minus optimized hybrid RPS; paired intervals condition on nine fixed old combinations. All four Holm-adjusted $p = .0008$. N=600.

| Features | Neutral comparator | Gain [paired 95% CI] |
|---|---|---|
| Categorical | Prior-anchored | 0.001465 [0.001213, 0.001722] |
| Categorical | Same ordinary ridge | 0.001843 [0.001595, 0.002088] |
| Raw | Prior-anchored | 0.001251 [0.001016, 0.001490] |
| Raw | Same ordinary ridge | 0.001747 [0.001519, 0.001974] |

## 5.7 Same-panel B1: fixed hybrid versus selected baselines

Figure 1 reports every frozen fitting budget. The two prespecified n=8,320 primary differences are *against* the hybrid: +0.021679 RPS for raw history (conditional paired 95% CI [0.020556, 0.022857]) and +0.018311 for the categorical profile ([0.017188, 0.019426]). Holm-adjusted $p = 0.0004$ for each, but both practical-benefit gates are false because their direction is positive. At n=80 the fixed hybrid scores better in both representations under its larger historical supervision. At n=160 raw is inconclusive (CI crosses zero) while categorical favors the hybrid; from n=320 every examined point favors the selected baseline. These are secondary, exposed-panel contrasts, not an estimated first crossing or a literal questionnaire answer-equivalent. The earlier full-8,000 ridge RPS 0.10995 versus raw hybrid 0.13153 already showed the broad gap; the new same-panel, within-budget CV family comparison tests that conclusion with capacity selection and both representations.

The accompanying aggregate CSV reports all twelve signed differences, intervals and adjusted tests. The same rows and their conditional estimand appear in Supplement S3, Table 1.

**How many observed answers match the fixed hybrid?** We held the 8,320 training respondents and the 60 target items fixed. For each of three frozen orders of the 60 input items we fitted statistical predictors to nested prefixes of 0, 1, 2, 5, 10, 20, 30, 40 and 60 answers. Three-fold training-respondent cross-validation selected the predictor at each prefix. We scored all predictors on the same exposed E3 panel of 600 people. The reference RPS, 0.131532, averages losses across the nine fixed historical raw-hybrid fits, each using all 60 input answers.

The decoder's mean RPS across orders decreased from 0.151894 with zero answers to 0.132718 with ten, 0.124497 with twenty and 0.109854 with all sixty. Every order yielded a monotone raw curve. Linear interpolation between ten and twenty answers gives $m^* = 11.44$ for the average curve. All 2,000 paired respondent-bootstrap replicates crossed within the grid (95% interval [10.46, 12.38], conditional on the fixed fits and item orders). Individual orders crossed at 9.17, 10.99 and 14.19 answers. Jointly resampling respondents and the three observed orders gives a descriptive interval of [9.13, 13.88]. The crossing mea-

sures equivalent predictive loss for this decoder and target. Prospective item selection and validation in new people would be needed to justify a shorter questionnaire.

**Equal-weight diagnostic on the exposed E3 panel.** At weight 0.5 with the same ordinary ridge, the old calibrated optimized forecast package beats its neutral counterpart by 0.001477 RPS for categorical features (paired 95% CI $[0.001291, 0.001676]$) and 0.001448 for raw history ($[0.001264, 0.001643]$); both Holm-adjusted $p = .0004$. All three historical-seed and all leave-one-seed-out mean differences are positive. This post hoc readback removes the E3 mixture-weight difference but preserves arm-specific calibration and reuses the same exposed people; it is neither a prompt-only effect nor fresh confirmation. The uncalibrated half-weight gains are larger but both absolute scores are worse. Full grid, controls, and independent raw-loss verification appear in Supplement S4.

## 5.8 External instrument: held-out calibration of the statistical control

In an unregistered auxiliary analysis, we evaluated the frozen R2 statistical control on the 200 calibration respondents excluded from its 1,200-person fit. Each respondent supplied one observed answer for each of the 49 endpoints. For each canonical product, selection within the training data considered the marginal prior, the posted price, the permitted history and the two sources combined. All 40 selected models used the actual posted price (21 price-only and 19 price-plus-history; 18 logistic and 22 tree models). The comparator is the product-specific, Laplace-smoothed response prior fitted on those same 1,200 people.

Across the 40 hypothetical purchase choices, mean log loss fell from 0.637315 (prior) to 0.519243 (selected decoder). The paired difference was $-0.118072$ (descriptive 95% respondent-bootstrap interval $[-0.133371, -0.102856]$); 39 products improved and one worsened. This contrast measures the fitted predictor's performance against a product-specific prior. Isolating the contribution of personal history requires a direct comparison with a price-only forecast.

Across the eight ordinal ratings, mean RPS was 0.121872 for the decoder and 0.121223 for the prior (difference $+0.000649$, 95% interval $[-0.001265, 0.002567]$); four ratings improved, three worsened and one tied. The marble-choice decoder selected the prior, with log loss 0.643622 for both. Intervals resample 200 people, conditional on the fitted models, averaging endpoints of the same type within person. The 1,400-person fit includes CAL and is reserved for the registered TEST comparison.

In the earlier, slot-indexed analysis, a history-only decoder selected the marginal forecast on 45 of 49 endpoints. At the 40 price slots, decoder and prior both scored 0.684166. Each slot pooled different products and prices across people. The R2 product-indexed scores use different target definitions, so a numerical difference between these two analyses cannot estimate an effect of the repair.

**Feasibility of stimulus-matched model forecasts.** On ten development respondents, Granite and Qwen each produced all 980 probability vectors (49 endpoints, two instructions). Across the 980 calls per model, the minimum total probability mass on valid answer tokens before renormalization was 0.9954 for Granite and 0.9716 for Qwen, above the frozen 0.5 threshold. Each model used one H200, for 56 min 24 s and 45 min 1 s respectively, processing 24.98 and 24.86 million input tokens. These checks use no outcome labels. They admitted both models to the 200-person calibration stage.

**Held-out prediction on the external survey.** All 658 TEST respondents had labels for all 49 endpoints. Relative to the statistical decoder fitted on 1,400 respondents, standalone LLMs and fixed half-weight mixtures had higher normalized loss in all eight comparisons. Both CAL-selected Granite mixtures improved modestly: $\Delta = -0.00404$ for neutral (95% CI $[-0.00723, -0.00096]$, Holm $p = .0360$) and $-0.00384$ for T0 ($[-0.00692, -0.00085]$, Holm $p = .0486$). Both CAL-selected Qwen mixture intervals included zero. In a post hoc direct paired comparison of the two Granite mixtures, the same CAL rule gave T0 minus neutral $\Delta = +0.000197$ (95% CI $[-0.000541, 0.000957]$). Both arms used the same calibration respondents, weight grid, loss and tie rule, with weights selected separately. At a common fixed weight of 0.5, the contrast was $-0.014990$ (95% CI $[-0.017155, -0.012732]$). The fixed-weight comparison favors T0. The CAL-selected contrast was near zero and its paired interval included zero. These exploratory contrasts reuse R2 TEST and are reported separately from the registered Holm families. In the standalone comparisons, T0 versus neutral reduced Granite loss by 0.05703 but increased Qwen loss by 0.03066. These are dimensionless differences normalized by endpoint-specific CAL prior loss, with equal weight for the two endpoint families. Intervals condition on the frozen forecasts. The complete results are in Supplement S3. In a separate exploratory decomposition on the same R2 TEST, the frozen CAL-selected Granite mixtures also improved on their own decoder1200 component: neutral $-0.002886$ (95% interval $[-0.004478, -0.001282]$), T0 $-0.002689$ ($[-0.004239, -0.001123]$). Both Qwen intervals included zero. Decoder1200 minus Full1400 was $-0.001150$ ($[-0.004417, 0.002007]$). Intervals are descriptive, conditional on fixed forecasts and outside the registered Holm families. The comparison holds the fitted decoder1200 fixed. The mixtures additionally use 200 CAL respondents for weight selection and require LLM forecasts. Full1400 fits on those CAL respondents as well as the original 1,200. The separate neutral contrast tests the two instruction-specific mixtures.

In the separate exploratory comparison on the 40 pric-

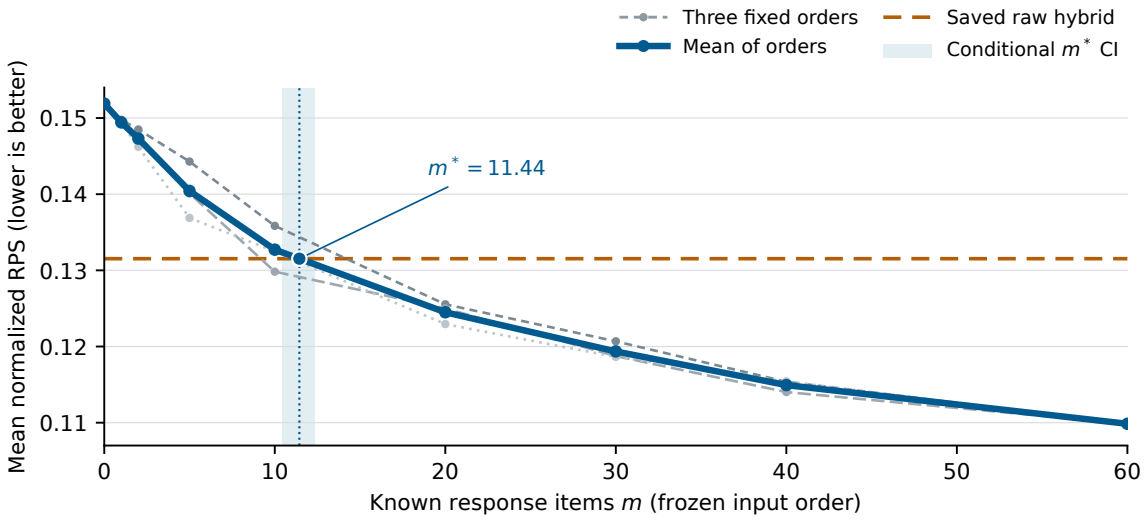



Figure 3: Observed answers needed to match the fixed hybrid on the exposed E3 panel. Thin lines show three frozen orders of input items. The thick line averages the decoder's losses across orders at each measured budget. The horizontal line marks the mean RPS of the nine historical raw-hybrid fits. The crossing interval resamples respondents, conditional on the three orders and fitted models.

ing products, adding the respondent's history to price increased mean log loss by 0.00420 (95% paired respondent interval $[-0.00015, 0.00852]$). The interval includes zero. This comparison uses the same 658 people and fixed statistical fits.

### 5.9 RQ3: Profile matching and prediction

The primary template instantiated with the matched profile outperforms the same template instantiated with a deranged respondent profile by 0.04278 RPS on Qwen and 0.03225 on Granite (Table 2). This comparison detects predictive signal from matching the supplied profile to its respondent; it does not test unique identification. It also does not isolate optimization, because neutral templates receive matched histories too.

The selection-chosen ridge predictor has held-out RPS 0.10955, compared with 0.13132 and 0.13594 for the primary Qwen and Granite personas. Ridge also outperforms both calibrated-fitness variants. For descriptive context, the prior scores 0.15174 and ExtraTrees 0.13127; evaluation scores are not used to change the selected decoder. The ridge comparison concerns the full prediction pipelines, given their different representations and fitting procedures.

Ridge using categorical profiles scores 0.11384; raw-history ridge scores 0.10955. Categorical-profile ridge retains 89.8% of raw-history ridge's prior-relative prediction gain. Every LLM arm is worse than categorical ridge in the 33-test family (all 28 relevant intervals positive, Holm $p < .01$). This is an observed prediction-gain ratio, not a fraction of human information or a causal compression decomposition. Ridge uses 8,000 labelled person-level mappings; the LLM uses those labels for priors plus search, selection and calibration targets. Supervision and learning procedures are not matched. Full representation contrasts and base-to-ridge descriptive ratios are in Supplement S4.

### 5.10 Exploratory budget and mixture diagnostics

On the exposed original cohort, calibrated GA can beat shared-prior ridge at 80 fitting people, while ridge has lower mean RPS at 320. Against raw-history ridge on exact Search80, Qwen gains 0.00530 RPS (crossed interval $[0.00422, 0.00638]$, conditional Holm $p = .0096$); Granite does not pass the combined rule. These fitting counts omit selection, calibration and prior supervision. Archival Part B learning curves and answer-equivalent diagnostics retain all budgets and their conditional uncertainty.

Mixtures beat both components in all 12 original conditions, but neutral controls restrict optimized-recipe utility. Against neutral plus anchored ridge, only Granite categorical/raw 80 pass the practical/statistical rule; Qwen loses in all six conditions, five significantly. This adaptive 24-test screen motivated E3, not a general optimizer claim. Full contrasts, weights, component gains and the separate 48-test family remain in Supplement S4.

## 6 Discussion

E1 supports historical calibrated Granite templates over neutral, but its MAP objective interval crosses zero; Qwen base advantages remain unsupported. Calibration changes

ranks and most same-shortlist winners without a clear selector benefit. Retaining historical winners precludes end-to-end optimizer replication or uniform transfer claims.

The third-panel Granite-80 recipe beats both neutral controls in categorical and raw representations. The ordinary-ridge control fixes coefficients but changes the old-selected weight with the prompt, supporting the recipe rather than a prompt-only or new-search effect. E4's smaller positive equal-weight package difference rules out unequal weights alone; it remains post hoc with distinct calibration.

The B1 comparison changes the practical interpretation. With the same 60 observed test-person items and a separately tuned statistical capacity at each fit budget, both baseline representations overtake the fixed hybrid by n=320 and are much better at n=8,320. The small-budget advantage of the hybrid survives as an observed point, with broader historical supervision than the baseline fit budget counts. These data do not identify the causal value of prompt text, nor a literal reduction in questions asked. A categorical-profile baseline also wins at large n, so uncompressed raw inputs alone cannot explain the full-data gap. Which component of the old mixture helps at small n remains a narrower question than whether a persona adds value over the strongest available same-history predictor.

# 7 Limitations and reproducibility

A separate calibration-only replay with pinned inputs, sources and serving settings failed a fixed numerical compatibility rule on one of two Qwen repeats; both Granite repeats passed (Supplement S4). This is not an optimizer test or a diagnosis of the cause of variation. Two subsequent full selected-template replays retain the GA objective contrast on this exposed cohort: raw-minus-calibrated RPS is 0.00161/0.00159 for Qwen and 0.00130/0.00130 for Granite. All four crossed intervals exclude zero (within-family Holm $p = 0.0008$; Supplement S4). This supports the contrast's robustness in these executions, but does not repair pointwise compatibility or establish generalization over serving environments. The selector study's strict selection audit failed on a post-receipt server teardown diagnostic; a versioned gate accepted that diagnostic while verifying the saved selection arrays. The final gate and independent CPU readback passed. Fresh Qwen neutral-base forecasts nevertheless have per-cell mean absolute drift of about 0.00031–0.00037 RPS from the archived base (maximum individual drift 0.0028–0.0048); Granite drift is much smaller. These checks do not repair pointwise serving compatibility. The planned E5 replication with new optimizer seeds remains unexecuted: its Granite readout failed the unchanged $5 \times 10^{-4}$ pointwise gate against E5's frozen calibrated-GA reference.

Evaluation covers one inventory, two readout families and existing-corpus participants; public-data pretraining exposure is unknown. E1 and E3 are disjoint from each other and the reachable inventoried exposure union, but share a corpus and item family; global lifetime exposure and untouched measurement choices remain unproven. Their targets did not tune the frozen recipes. Transfer fixes old prompts and calibration, so it does not test new optimizer runs, alternate selector winners or external behavior.

E3 fixes three old search seeds, three ridge subsets and old-selected weights; its intervals condition on those choices. Its protocol followed earlier-outcome inspection. Neither fit-$n = 80$ nor a new respondent panel implies equal total supervision or external population validation. E4 reuses E3's exposed labels and forecasts, adding neither a second transfer panel nor independent old-seed replication. Other post hoc specifications preceded execution but their questions followed original outcomes. Three seeds limit inference over optimizer randomness; pointwise bootstrap intervals are not simultaneous across methods.

B1 reuses the now-exposed E3 panel and averages nine correlated old hybrid combinations against three fixed baseline subsets. Its respondent intervals are conditional on these runs; Holm-adjusted sign-flip tests do not turn this into a fresh confirmation. The n=8,320 contrast compares total unique task-labelled people touched somewhere in the historical LLM recipe to a statistical fitting budget, not identical supervision semantics or compute. The Twin-2K-500 contrasts condition on the two fixed model revisions, instructions, fitted decoders and CAL weights. R2 was frozen after inspection of the earlier history-only CAL diagnostic but before its own fitting and GPU forecasts. Its exact-stimulus repair resolves a measurement mismatch, not the risk of model pretraining exposure or post-CAL design influence. Wave-4 answers are public, and hypothetical choices/ratings do not validate real purchases or behavior.

Equal forecast budgets do not imply equal total cost. The recorded historical historical minimum is 300.60 allocated GPU-hours. Failed/cancelled request work, other development, mutator/completion tokens, CPU work and dollars remain unresolved. The separate replays omit search, prior fitting and old calibration costs; forecast/token/allocation accounting appears in Supplement S4 (Tables S15–S16, combined PDF p. 18). No end-to-end efficiency claim follows. Native GigaEvo is private: reproduction requires access or a separately validated substitute, which must not be labelled native reproduction.

Manifests, model locks, templates, probabilities, calibration and loss receipts are retained. Human histories and exposure IDs require de-identification and source permissions; labelled training humans count toward cost. Shared data/methodology do not establish an independent publication contribution.

**AI assistance.** This revision additionally used Claude Code for independent manuscript critique.

Codex assisted inspection, retrieval, design, analysis and writing; Claude supplied generic plotting/layout designs. Scientific labels and measurements were bound

locally; figures were checked against recorded outputs. Supplement S4 records available versions, prompts and historical-provenance limits.

# 8 Conclusion

The frozen Granite calibrated-GA persona-plus-ridge recipe beats two neutral mixtures on a third, same-corpus panel. Its practical comparison with a capacity-selected same-history predictor is less favorable: at 8,320 fitted people the hybrid has 0.021679 higher raw-history RPS and 0.018311 higher profile-history RPS, with both declared practical gates false. Both baseline representations lead from 320 fitted people onward; the low-budget hybrid advantage is conditional on its old, larger supervision pipeline. With training size fixed at 8,320, a raw-history decoder reaches the 60-answer hybrid's mean loss at an interpolated 11.44 known answers on the same panel. The original planned optimizer comparison is null; E1's Granite neutral gains and E2/E4 diagnostics retain their narrower interpretations. We therefore measure a boundary on persona value, rather than claim general superiority, sample-efficiency savings, or behavioral transfer. The external-protocol validity audit recovered each respondent's randomized product and price and bound both model arms and controls to that same stimulus. In that survey, adaptive mixing gave a small Granite gain over the full-data control for both instructions. The opposing standalone T0 effects across Granite and Qwen limit any claim of portable persona benefit. The direct CAL-selected Granite comparison yielded a near-zero estimate (Section 5.8).

# References


Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective prompt evolution can outperform reinforcement learning. In *International Conference on Learning Representations*, 2026. URL `https://arxiv.org/abs/2507.19457`.

Gati V. Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 337–371, 2023. URL `https://proceedings.mlr.press/v202/aher23a.html`.

Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023. doi: 10.1017/pan.2023.2.

Farima Fatahi Bayat, Moin Aminnaseri, Pouya Pezeshkpour, and Estevam Hruschka. Reflective prompt tuning through language model function-calling, 2026. URL `https://arxiv.org/abs/2605.21781`.

Marcel Binz, Elif Akata, Matthias Bethge, Franziska Brändle, Fred Callaway, Julian Coda-Forno, Peter Dayan, Can Demircan, Maria K. Eckstein, Noémi Éltető, Thomas L. Griffiths, Susanne Haridi, Akshay K. Jagadish, Li Ji-An, Alexander Kipnis, Sreejan Kumar, Tobias Ludwig, Marvin Mathony, Marcelo Mattar, Alireza Modirshanechi, Surabhi S. Nath, Joshua C. Peterson, Milena Rmus, Evan M. Russek, Tankred Saanum, Johannes A. Schubert, Luca M. Schulze Buschoff, Nishad Singhi, Xin Sui, Mirko Thalmann, Fabian J. Theis, Vuong Truong, Vishaal Udandarao, Konstantinos Voudouris, Robert Wilson, Kristin Witte, Shuchen Wu, Dirk U. Wulff, Huadong Xiong, and Eric Schulz. A foundation model to predict and capture human cognition. *Nature*, 644:1002–1009, 2025. doi: 10.1038/s41586-025-09215-4.

James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. Synthetic replacements for human survey data? the perils of large language models. *Political Analysis*, 32(4):401–416, 2024. doi: 10.1017/pan.2024.5.

Chrisantha Fernando, Dylan Sunil Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: Self-referential self-improvement via prompt evolution. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 13481–13544. PMLR, 2024. URL `https://proceedings.mlr.press/v235/fernando24a.html`.

Leon Fröhling, Jens Rupprecht, Markus Strohmaier, and Claudia Wagner. When persona attributes improve population alignment in large language models, 2026. URL `https://arxiv.org/abs/2609.02526`.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Tilmann Gneiting and Roopesh Ranjan. Combining predictive distributions. *Electronic Journal of Statistics*, 7:1747–1782, 2013. doi: 10.1214/13-EJS823. URL `https://doi.org/10.1214/13-EJS823`.

Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In *International Conference on Learning*

*Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/9156b0f6dfa9bbd18c79cc459ef5d61c-Abstract-Conference.html.

Tiancheng Hu and Nigel Collier. Quantifying the persona effect in LLM simulations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10289–10307, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.554. URL https://aclanthology.org/2024.acl-long.554/.

Nusrath Jahan and Johnathan Mell. Unraveling the tapestry of deception and personality: A deep dive into multi-issue human-agent negotiation dynamics. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*, pages 916–925. IFAAMAS, 2024. doi: 10.65109/sjpk9444. URL https://www.ifaamas.org/Proceedings/aamas2024/pdfs/p916.pdf.

Mumin Jia, Yilin Chen, Divya Sharma, and Jairo Diaz-Rodriguez. When can digital personas reliably approximate human survey findings?, 2026. URL https://arxiv.org/abs/2605.10659v3. Accepted at NeurIPS 2026; arXiv version 3, 13 October 2026.

John A. Johnson. Measuring thirty facets of the five factor model with a 120-item public domain inventory: Development of the IPIP-NEO-120. *Journal of Research in Personality*, 51:78–89, 2014. doi: 10.1016/j.jrp.2014.05.003. URL https://doi.org/10.1016/j.jrp.2014.05.003.

Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites, 2015. URL https://arxiv.org/abs/1504.04909.

Lewis Newsham and Daniel Prince. Personality-driven decision-making in LLM-based autonomous agents. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems*, pages 1538–1547. IFAAMAS, 2025. URL https://ifaamas.org/Proceedings/aamas2025/pdfs/p1538.pdf.

Joon Sung Park, Carolyn Q. Zou, Jonne Kamphorst, Niles Egan, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Percy Liang, Robb Willer, and Michael S. Bernstein. LLM agents grounded in self-reports enable general-purpose simulation of individuals, 2026. URL https://arxiv.org/abs/2411.10109v3. Version 3, 28 June 2026; first posted in 2024.

Reid Pryzant, Dan Iter, Jerry Li, Yin Tat Lee, Chenguang Zhu, and Michael Zeng. Automatic prompt optimization with "gradient descent" and beam search. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7957–7968. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-main.494. URL https://aclanthology.org/2023.emnlp-main.494/.

Mikayel Samvelyan, Davide Paglieri, Minqi Jiang, Jack Parker-Holder, and Tim Rocktäschel. Multi-agent diagnostics for robustness via illuminated diversity. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*, pages 1630–1644. IFAAMAS, 2024. doi: 10.65109/OPQC5611. URL https://www.ifaamas.org/Proceedings/aamas2024/pdfs/p1630.pdf.

Lindia Tjuatja, Valerie Chen, Tongshuang Wu, Ameet Talwalkwar, and Graham Neubig. Do LLMs exhibit human-like response biases? a case study in survey design. *Transactions of the Association for Computational Linguistics*, 12:1011–1026, 2024. doi: 10.1162/tacl_a_00685. URL https://aclanthology.org/2024.tacl-1.56/.

Olivier Toubia, George Z. Gui, Tianyi Peng, Daniel J. Merlau, Ang Li, and Haozhe Chen. Database report: Twin-2k-500: A data set for building digital twins of over 2,000 people based on their answers to over 500 questions. *Marketing Science*, 44(6):1446–1455, 2025. doi: 10.1287/mksc.2025.0262. URL https://doi.org/10.1287/mksc.2025.0262.

Weilai Xu, Fred Charles, and Charlie Hargood. Generating stylistic and personalized dialogues for virtual agents in narratives. In *Proceedings of the 22nd International Conference on Autonomous Agents and Multiagent Systems*, pages 737–746. IFAAMAS, 2023. URL https://www.ifaamas.org/Proceedings/aamas2023/pdfs/p737.pdf.

Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/3339f19c5fcee3ad74502947a32be9e6-Abstract-Conference.html.

# Supplement S3: External protocol and budget results to "What Does a Persona Add?"

Evidence lineage, paired capacity comparison, and external stimulus protocol

Anonymous authors

## 1 Scope and source lineage

This supplement records the retrospective B1 capacity comparison and R2 stimulus-matched protocol and results. The historical diagnostics are supplied separately as Supplement S4. Protocol freezing is internal, not public preregistration.

The external source is `LLM-Digital-Twin/Twin-2K-500`. Its pinned revision is `f883165a3026fde855dfd448e0cd16443ab257b6`

The original 49-endpoint catalog protocol is `TWIN_PROTOCOL_FROZEN.json`. Its file SHA-256 is `0ea4652877ccc93b288e71172fb5a4d473985c431b794fb095434746622fae17`

The respondent-specific R2 amendment is `TWIN_STIMULUS_AMENDMENT.json`. Its SHA-256 is `1a968951bbf7f11a5d9a838b00676e058de1896a7778bece04ba0b1d3eb70067`

It binds the parent protocol, prepared receipt, question-only sidecar, FIT orientation audit, canonical target-count ledger, and presentation-slot response semantics. The source and FIT-only receipts are the `source audit` and `FIT orientation audit`. The R2 amendment was frozen *after* the earlier v6 history-only CAL readback but *before* any R2 fit or GPU forecast. All 84 compute cluster source tests and complete R2 input validation are engineering checks, not predictive results.

## 2 B1: complete paired capacity comparison

The target is normalized ordinal ranked probability score (RPS) on 60 disjoint target items for each of the same 600 E3 people. For $K$ categories,

$$\mathrm{RPS}(p, y) = \frac{1}{K-1} \sum_{k=1}^{K-1} \left( \sum_{j=1}^{k} p_j - \mathbf{1}\{y \le k\} \right)^2 .$$

Lower is better. The contrast is the archived E3 frozen hybrid minus the within-budget, cross-validated statistical predictor. The hybrid's per-person loss is averaged across nine fixed correlated combinations (three old LLM seeds times three ridge subset seeds); the reference's per-person loss is averaged across three fixed subset seeds. Forecast probabilities are scored before averaging losses. The nine historical combinations are not nine independent optimizer runs. The 600 respondent differences, not individual item answers, are the inferential units.

At each budget $n \in \{80, 160, 320, 1000, 8000, 8320\}$ and in each of the raw-answer and exact categorical-profile representations, the frozen candidate families were an empirical prior, ridge penalties $\{1, 10, 100, 1000\}$, and a small nonlinear kNN/ExtraTrees family. Within each $n$ and each of three nested subset seeds, three-fold training-only cross-validation selected family and hyperparameters by mean held-person RPS, with registered ties. The $8,320$ pool comprises the historical Train8000, Search80, Select120 and Calibrate120 people; no E3 Eval600 label entered fitting or selection. All 18 budget-by-seed selections and 108 family forecasts were retained. The historical hybrid itself nevertheless used additional search, prior, selection and calibration supervision, so $n$ is a *statistical fit budget*, not total unique human information or literal questionnaire answers saved.

Table 1: All 12 retained B1 contrasts. $\Delta$ is hybrid minus selected reference normalized RPS; positive favors the reference. Intervals resample 600 respondents conditional on fixed forecasts. $p_{H,12}$ is the separate 12-contrast secondary Holm family; the two $n = 8320$ rows have primary Holm-adjusted $p_{H,2} = 0.0004$.

| $n$ | History | $\Delta$ | Conditional 95% CI | $p_{H,12}$ |
|---|---|---|---|---|
| 80 | Raw | −0.007086 | [−0.007705, −0.006471] | 0.0024 |
| 160 | Raw | −0.000500 | [−0.001047, 0.000066] | 0.0820 |
| 320 | Raw | 0.005355 | [0.004644, 0.006076] | 0.0024 |
| 1000 | Raw | 0.012844 | [0.011920, 0.013785] | 0.0024 |
| 8000 | Raw | 0.021580 | [0.020406, 0.022741] | 0.0024 |
| 8320 | Raw | 0.021679 | [0.020556, 0.022857] | 0.0024 |
| 80 | Categorical | −0.008770 | [−0.009444, −0.008102] | 0.0024 |
| 160 | Categorical | −0.002250 | [−0.002824, −0.001697] | 0.0024 |
| 320 | Categorical | 0.004270 | [0.003631, 0.004925] | 0.0024 |
| 1000 | Categorical | 0.011452 | [0.010559, 0.012307] | 0.0024 |
| 8000 | Categorical | 0.018254 | [0.017158, 0.019373] | 0.0024 |
| 8320 | Categorical | 0.018311 | [0.017188, 0.019426] | 0.0024 |

The receipt records 5,000 respondent bootstrap draws and 5,000 paired two-sided sign flips. Both $n = 8320$ primary contrasts have Holm-adjusted $p = 0.0004$, but neither passes the *benefit* gate: $\Delta \leq -0.001$, upper CI below $-0.001$, and Holm $p < 0.05$ were all required together. At $n = 80$ the hybrid leads despite its hidden historical supervision; $n = 160$ differs by representation; from $n = 320$ both selected references lead. These are conditional estimates on the already exposed E3 panel.

# 3 R2 respondent-specific stimulus crosswalk

The original catalog stores the first question it saw for a QID, while the survey presented 40 product categories in randomized slots and with respondent-specific prices. A question-only source scan read the respondent ID and source question objects, not the Wave-4 numeric answer column. For each respondent it masked exactly one posted price, matched the remaining text to a canonical product, preserved that product's actual price and ordered options, and checked that every product appeared once. The nine nonpricing target texts and options were invariant. The full ledger is:

| Verified item | Count |
|---|---|
| Respondents / canonical endpoints | 2,058 / 49 |
| Product offers / products per respondent | 82,320 / 40 |
| Product-slot pairs observed | 1,600 |
| Price levels per product / zero-price offers | 11 / 7,490 |
| Offers exactly matching old catalog example | 220 |
| Nonpricing endpoints with invariant text/options | 9 |
| FIT-only question-text matches | 43,000 / 43,000 |
| FIT-only numeric slot-choice agreements | 49,000 / 49,000 |

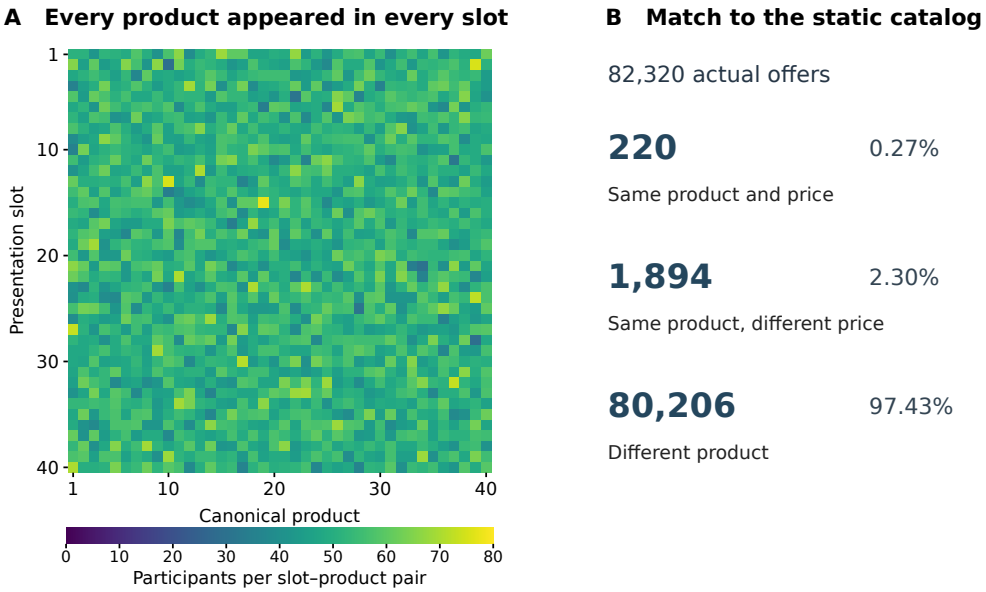



Figure 1: Outcome-blind product-by-presentation-slot counts (left) and comparison with the static catalog example (right). Of 82,320 actual offers, 220 matched both catalog product and price, 1,894 matched product but not price, and 80,206 displayed a different product in the catalog's slot. Counts derive from question text and price only; no participant answer enters the figure.

The numeric orientation audit inspected only the 1,000 frozen FIT people to establish that `QID9_j` numeric response columns index randomized *presentation slots*. Its 49,000 agreements connect selected option positions in the source question object to the numeric export. DEV, CAL and TEST answers were not inspected for this orientation check; no predictive model was fit or scored by it. R2 maps a raw slot answer to a canonical product answer using the verified question-only slot/product permutation. Both LLM arms receive the exact respondent question and both statistical controls receive the same product/price/condition fields. The sidecar, amendment, option order, counts and source code are hash-bound at input load and comparison. The earlier v6 history-only CAL result (prior chosen for 45 of 49 endpoints) marginalizes over this offer variation and does not measure whether history adds value conditional on a known offer.

# 4 External endpoints, scoring, and primary aggregation

The frozen endpoint definition is in `TWIN_PROTOCOL_FROZEN.json`; the following table fixes its groups and scales without disclosing participant data.

| Family | Canonical columns | $K$ | Primary score |
|---|---|---|---|
| Economic preferences | `QID9_1`-`QID9_40`, 40 canonical products | 2 | Exact log loss |
| Heuristics/cognitive bias | `QID288_1-4`, `QID289_1-4`, eight ratings | 7 | Normalized RPS |
| Heuristics/cognitive bias | `QID196`, one binary choice | 2 | Exact log loss |

All probabilities must match every respondent, endpoint, ordered option and split. Missing human labels use one fixed observed mask per endpoint across all methods; missing *forecasts* fail the panel. For nominal/binary endpoints, $L(p, y) = -\log p_y$ without clipping; $p_y = 0$ yields infinite loss and unavailable aggregate inference. Brier is secondary. For the eight ordinal tasks, $L$ is the normalized RPS above with $K = 7$.

The frozen primary overall estimand is dimensionless. Let $D_e$ be the mean CAL primary loss of the FIT1000-only Laplace prior with $\alpha = 1$ for endpoint $e$ (zero history). For a system $S$ and reference $R$, let

$$\delta_e = \frac{1}{|O_e|} \sum_{i \in O_e} \frac{L_{ie}(S) - L_{ie}(R)}{D_e}, \qquad \Delta = \frac{1}{2}\left(\frac{1}{40} \sum_{e \in E_{40}} \delta_e\right) + \frac{1}{2}\left(\frac{1}{9} \sum_{e \in H_9} \delta_e\right).$$

Here $O_e$ is the fixed set of observed TEST labels at endpoint $e$. The freeze records each prior vector, FIT and CAL labeled counts, and $D_e$; $D_e \leq 10^{-6}$ or no CAL labels aborts. Every endpoint needs at least 30 observed TEST labels

or the *entire* primary aggregate is unavailable. This averages normalized *differences* within endpoint, then endpoints within family, then the two families equally; it never pools raw log loss and RPS. The three-subfamily variant was explicitly not chosen as primary. A respondent bootstrap or sign vector is shared across all 49 endpoints, retaining within-person dependence. Empty-mask bootstrap draws are counted and the valid-draw conditional CI and failure fraction reported. Holm correction covers the complete 12 primary system-versus-full1400 contrasts separately from the two T0-versus-neutral contrasts; endpoint tables use a separate secondary Holm family.

# 5 Frozen model, comparator, and failure gates

The R2 statistical family includes an empirical Laplace prior and predeclared regularized logistic and ExtraTrees candidates using history, respondent-specific stimulus, or both. Three-fold CV inside the 1,200 FIT+DEV people is specified to choose each endpoint's family and settings before CAL or TEST scoring. The `decoder1200` CAL forecast must be held out. The `full1400` comparator is specified to refit using FIT+DEV+CAL and to be scored only on TEST; its in-sample CAL output is ineligible for mixture tuning.

The two fixed LLM arms are neutral and the portable T0 instruction component, on each pinned model. The instruction is not the full old NEO genotype. A CPU preflight binds source hashes, model/tokenizer commits, architecture, exact complete prompt tokenization and a no-truncation context ceiling. A finite one-GPU DEV probe tests 10 people across every endpoint and both arms before CAL. Each forecast is the exact full-softmax mass on registered canonical digit tokens, normalized within that set; the frozen minimum digit-mass gate must pass. CAL alone is designated to choose, per arm and endpoint, $w \in \{0, 0.1, \ldots, 1\}$ for $wP_{\mathrm{LLM}} + (1-w)P_{1200}$ by the endpoint's primary score. The fixed $w = 0.5$ mixture is also retained. TEST starts only after the mixture freeze binds the R2 stimulus lineage, prepared source, options, and same model/readout. Direct arms, fixed mixtures and selected mixtures are all compared to full1400 without outcome-based deletion.

An incomplete forecast grid, mismatched labels or options, source-hash drift, failed digit-mass/context gate, nonfinite probability or unbound receipt is a failed or unavailable comparison, never an imputed zero effect. Exact-zero true-answer log loss remains infinite; no clipping or selective method dropout repairs it. Scores retain negative, mixed and failed rows. One model's engineering failure does not certify a scientific effect for the other. TEST remains sealed until the declared gates pass.

# 6 Resource and exposure ledger

Finite Slurm allocations are required for CPU prepare/selection/scoring and for single-GPU model inference. Receipts distinguish allocated CPU-hours, allocated GPU-hours, elapsed seconds, prompt tokens, attempted/completed forecasts, failed tasks, and unknown fields; an absent field is *unknown*, not zero. Both DEV models produced 980 forecasts; their combined GPU allocation was 1.69 hours. The pilot-based resource forecast for CAL and TEST was 218.52 GPU-hours across both models. The two CAL and two TEST GPU allocations used 143.89 hours in total. The historical minimum of 300.60 allocated GPU-hours excludes known-unsummed search/development and failure components, so it is not an end-to-end cost ratio against the fitted statistical controls. The external Wave-4 labels are public, model pretraining exposure is unknown, and the instrument measures hypothetical answers rather than purchases.

AI assistance is disclosed in the main manuscript.

# 7 Verified external TEST results

The separate numerical audit reconstructed all 14 primary and 686 endpoint contrasts, including the frozen respondent bootstrap and sign-flip draws. Negative values favor the system. The two contrasts against neutral form their own Holm family.

| System | Reference | Δ | 95% interval | Holm $p$ |
|---|---|---|---|---|
| Granite Neutral | Full1400 | 0.43267 | [0.41554, 0.44938] | 0.0024 |
| Granite Neutral half | Full1400 | 0.09614 | [0.08774, 0.10418] | 0.0024 |
| Granite Neutral CAL | Full1400 | -0.00404 | [-0.00723, -0.00096] | 0.0360 |
| Granite T0 | Full1400 | 0.37564 | [0.35879, 0.39205] | 0.0024 |
| Granite T0 half | Full1400 | 0.08115 | [0.07308, 0.08924] | 0.0024 |
| Granite T0 CAL | Full1400 | -0.00384 | [-0.00692, -0.00085] | 0.0486 |
| Granite T0 | Granite neutral | -0.05703 | [-0.06204, -0.05193] | 0.0004 |
| Qwen Neutral | Full1400 | 0.30872 | [0.29367, 0.32358] | 0.0024 |
| Qwen Neutral half | Full1400 | 0.06902 | [0.06117, 0.07686] | 0.0024 |
| Qwen Neutral CAL | Full1400 | -0.00202 | [-0.00503, 0.00106] | 0.4107 |
| Qwen T0 | Full1400 | 0.33939 | [0.32222, 0.35712] | 0.0024 |
| Qwen T0 half | Full1400 | 0.07664 | [0.06800, 0.08536] | 0.0024 |
| Qwen T0 CAL | Full1400 | -0.00110 | [-0.00427, 0.00211] | 0.5007 |
| Qwen T0 | Qwen neutral | 0.03066 | [0.02269, 0.03899] | 0.0004 |

On the 40 pricing products, the separate exploratory price-plus-history minus price-only contrast was +0.00420 mean log loss (paired respondent 95% interval $[-0.00015, 0.00852]$). All 26,320 paired cells were observed and finite.

**Exploratory direct comparison of Granite mixtures.** Both contrasts are T0 minus neutral on 658 people and all 49 endpoints, using the frozen R2 normalization and equal family weights. Intervals are paired respondent percentile bootstrap intervals (10,000 draws, seed 20261110). The CAL-selected contrast is +0.000197 ($[-0.000541, 0.000957]$). Weights differed on 18 endpoints; mean Granite weights were 0.1020 (neutral) and 0.0980 (T0). At a common weight of 0.5, the contrast is −0.014990 ($[-0.017155, -0.012732]$). These are post hoc contrasts outside the registered Holm families. The analysis read frozen forecasts.

**Exploratory mixture decomposition.** Each CAL-selected mixture is compared with its own decoder1200. The bridge is decoder1200 minus Full1400. All 658 respondents and 49 endpoints were reused. Descriptive paired respondent 95% intervals use 10,000 draws (seed 20261110), outside the registered Holm families. Negative values favor the system. Decoder1200 vs Full1400: −0.001150 ($[-0.004417, 0.002007]$). Granite neutral: −0.002886 ($[-0.004478, -0.001282]$). Granite T0: −0.002689 ($[-0.004239, -0.001123]$). Qwen neutral: −0.000872 ($[-0.003070, 0.001314]$). Qwen T0: +0.000050 ($[-0.001549, 0.001636]$).

# Supplement S4: Historical diagnostics

Anonymous authors

## 1 Source-data identity audit

A separate CPU audit compared the derived corpus with the original IPIP120.dat from Johnson's data repository. The article describes the IPIP-NEO-120 study Johnson [2014]; the later OSF deposit linked here supplies the audited source file. The original file contains 619,150 records. All 410,168 retained derived records matched exactly by case identifier and the 120 recoded item values: zero absent cases and zero item mismatches. The source file's SHA-256 was checked against its published OSF metadata, and the derived corpus's SHA-256 matched the frozen experiment input before and after the audit. Only aggregate counts were exported; respondent records used private temporary storage. The original DAT120 codebook confirms that reverse-keyed items were already recoded and zero denotes missingness. No second reversal was applied. This verifies retained case/item identity, not demographic-field equivalence, completeness of the historical filtering procedure, participant consent, upstream data permissions, or absence from pretraining. The audit did not estimate predictive performance or inspect panel-specific outcomes.

## 2 Scope

The evidence map in Table S3 distinguishes the analysis families. These additional analyses are post hoc, using the exposed original 600-person cohort and three search seeds. They do not supply a new confirmatory sample. Original 110-test results are unchanged. The initial representation tests use one 33-test Holm family; direct objective and interaction tests use a separate nine-test family. The figure/table differences are computed from unrounded values. Intervals for LLM comparisons cross respondents and observed search seeds; sign-flip p-values condition on those seeds. Ridge-only contrasts resample respondents with trained predictors held fixed.

**Objective contrasts.** The first four contrasts are raw-fitness minus calibrated-fitness RPS. Positive values favor calibrated fitness. Remaining rows subtract these paired objective effects; they are not inferred from subgroup significance. All five interaction intervals contain zero. No effect is claimed to generalize to an unseen optimizer seed. The smallest possible two-sided sign-test p-value with three seed pairs is .25; the table instead uses respondent-level sign flips after averaging the three seeds.

## 3 Representation controls

Categorical ridge uses 175 one-hot features: five categories for each of 30 facets and five traits. Continuous ridge uses the same 35 values before discretization, standardized on training people. Both use 8,000 training and 120 selection respondents, choosing penalty from $\{1, 10, 100, 1000\}$ before reading evaluation labels. Selected penalties are 100 and 1, respectively. Categorical, continuous and raw-history ridge score 0.11384, 0.11558 and 0.10955. Their pairwise differences (categorical–raw, continuous–raw, categorical–continuous) are 0.00428, 0.00603 and $-0.00174$; corresponding 95% intervals are $[.00378, .00478]$, $[.00540, .00664]$ and $[-.00231, -.00120]$. All have Holm $p = .006599$. These are fitted-model gaps, not a causal decomposition or an information bound.

## 4 Search ranking diagnostic

Each observed pool contributes the first complete evaluation per distinct candidate. We compare the saved raw and cross-fitted-calibrated scores, excluding tied pairs from the discordance denominator. The 24 pools span both models, three seeds, two optimizers and two objectives. Median Spearman correlation is .344, median discordant fraction .356, and median overlap of the two top-five sets is zero. These are within-search descriptors, not independent hypothesis tests or evidence of causal mediation.

**Independent summary recomputation.** A separate implementation reconstructed the 48 mixture-screen and 24 prior-anchored extension comparisons from the retained individual loss arrays. For all 72 contrasts it exactly reproduced the mean differences, 5,000-draw respondent-bootstrap intervals, conditional sign-flip probabilities, Holm adjustments and joint screen decisions. The same declared random stream was used for numerical comparison. This check validates aggregation from saved losses; it does not regenerate model forecasts, refit predictors, or provide independent confirmation of the adaptive scientific findings. The aggregate receipt and checking source accompany the evidence bundle; individual losses remain on the research host.

## 5 Artifact inventory

Machine-readable aggregates contain all 33 representation contrasts and all 9 objective contrasts, plus 24 history

Table S1: Raw and calibrated losses of the same selected templates; the final column lists three seed-specific calibrated differences from each cell's base.

| Model | Search family, fitness | Raw RPS | Calibrated RPS | Differences by seed |
|---|---|---|---|---|
| Qwen3.6 | GA, raw | 0.15665 | 0.13127 | +0.00132, +0.00134, +0.00087 |
| Qwen3.6 | GA, calibrated | 0.16377 | 0.12965 | -0.00066, +0.00003, -0.00069 |
| Qwen3.6 | MAP-Elites, raw | 0.15577 | 0.13132 | -0.00053, +0.00045, +0.00378 |
| Qwen3.6 | MAP-Elites, calibrated | 0.17042 | 0.12976 | -0.00029, -0.00087, +0.00015 |
| Granite | GA, raw | 0.17209 | 0.13520 | -0.00090, -0.00195, -0.00123 |
| Granite | GA, calibrated | 0.18219 | 0.13391 | -0.00295, -0.00280, -0.00223 |
| Granite | MAP-Elites, raw | 0.17640 | 0.13594 | -0.00202, -0.00096, +0.00110 |
| Granite | MAP-Elites, calibrated | 0.18436 | 0.13389 | -0.00203, -0.00385, -0.00214 |

Table S2: All nine post-hoc objective contrasts. Intervals cross respondents and observed seeds; Holm $p$ values condition on the three frozen runs.

| Contrast | Difference | Crossed 95% interval | Conditional $p_{\mathrm{H}}$ |
|---|---|---|---|
| Qwen:GA:raw–calibrated | +0.00163 | [+0.00111,+0.00219] | 0.002 |
| Qwen:MAP:raw–calibrated | +0.00156 | [-0.00012,+0.00347] | 0.002 |
| Granite:GA:raw–calibrated | +0.00130 | [+0.00060,+0.00204] | 0.002 |
| Granite:MAP:raw–calibrated | +0.00205 | [+0.00015,+0.00340] | 0.002 |
| Qwen: GA–MAP objective | +0.00007 | [-0.00190,+0.00206] | 0.719 |
| Granite: GA–MAP objective | -0.00075 | [-0.00249,+0.00185] | 0.002 |
| GA:Qwen–Granite | +0.00033 | [-0.00028,+0.00099] | 0.252 |
| MAP:Qwen–Granite | -0.00049 | [-0.00158,+0.00049] | 0.047 |
| Optimizer–model interaction | +0.00082 | [-0.00029,+0.00211] | 0.012 |

summaries. The fixed diagnostic protocol, analysis scripts and figure scripts accompany this local draft. Raw human records remain on the research host. Their inclusion in an anonymous release requires a separate permission and de-identification review; this local inventory is not a claim that a public release already exists.

Table S3: Distinct analysis families. Original post-hoc families reuse the exposed Evaluate600; fixed-template transfer uses 600 additional people from the same corpus. Holm corrections are within families, not across the adaptive research program.

| Question / family | Status | Target and interpretation |
|---|---|---|
| Primary and original arm contrasts / 110 | Frozen before evaluation | Respondent mean over three frozen runs; crossed intervals also resample observed runs. |
| Representation / 33 | Post hoc | Fitted-model comparison at 8,000 labels, same exposed people. |
| Objectives and interactions / 9 | Post hoc | Direct run-index paired effects, conditional sign flips; three seeds. |
| Ordinary budget / 48; shared-prior budget / 48 | Sequential adaptive diagnostics | Fixed fitted predictors at each budget, three LLM runs; separate families. |
| Mixture screen / 48 | Adaptive exploratory | Respondent effects conditional on nine fixed fit/run combinations; no optimizer or subset resampling. |
| Anchored rival / 24 | Subsequent adaptive extension | Frozen original hybrid versus prior-anchored ridge and its neutral mixture; same nine fixed combinations. |
| Readout replay / 4 intervals | Engineering diagnostic | Calibration-only numerical compatibility, separate from predictive improvement. |
| Selected-template replay / 4 primary + 4 secondary | Frozen replay on exposed data | Two fresh-server rounds of the GA objective contrast; separate Holm-four families. No new optimizer runs or people. |
| Additional-respondent transfer / 3 primary | Fixed after original outcomes | Historical GA/MAP winners on 600 inventoried-disjoint same-corpus people; three Granite tests, conditional on old seeds. Qwen and ridge are mandatory diagnostics. |
| Same-shortlist selector / 10 | Subsequent exploratory intervention | Five candidates from each of 12 archived GA pools; raw versus cross-fitted calibrated selection, then the old exposed Evaluate600. Four selector effects, four search effects, two interactions; three old seeds. |

# 6 Full method visualization

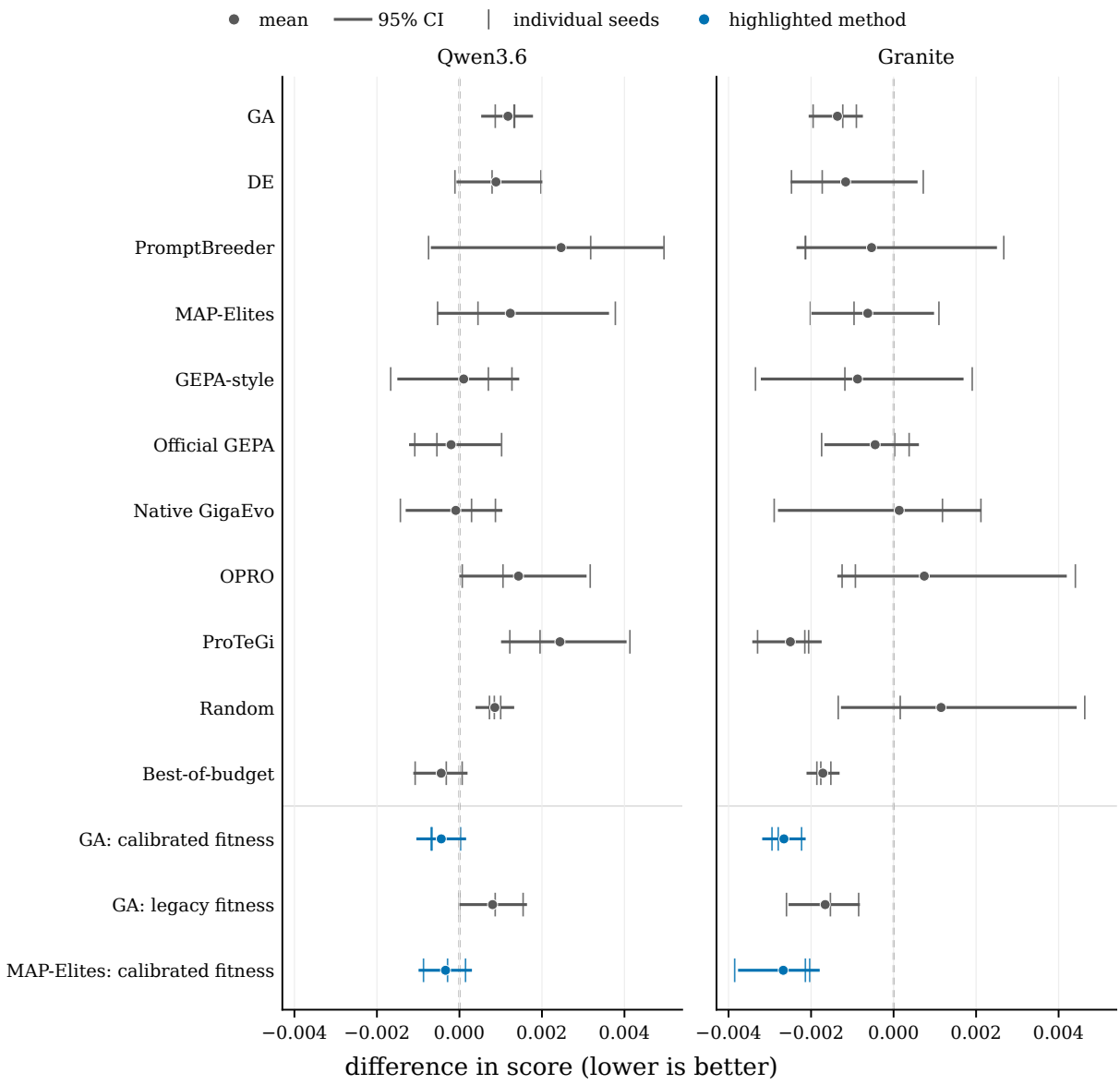



**Figure S1.** All 14 method/objective identities relative to their own base. Circles are mean differences, bars are crossed 95% intervals, and ticks are seed values. Blue highlights calibrated-fitness variants. The family remains the original 110 comparisons.

# 7 Paired MAP-Elites objective effects

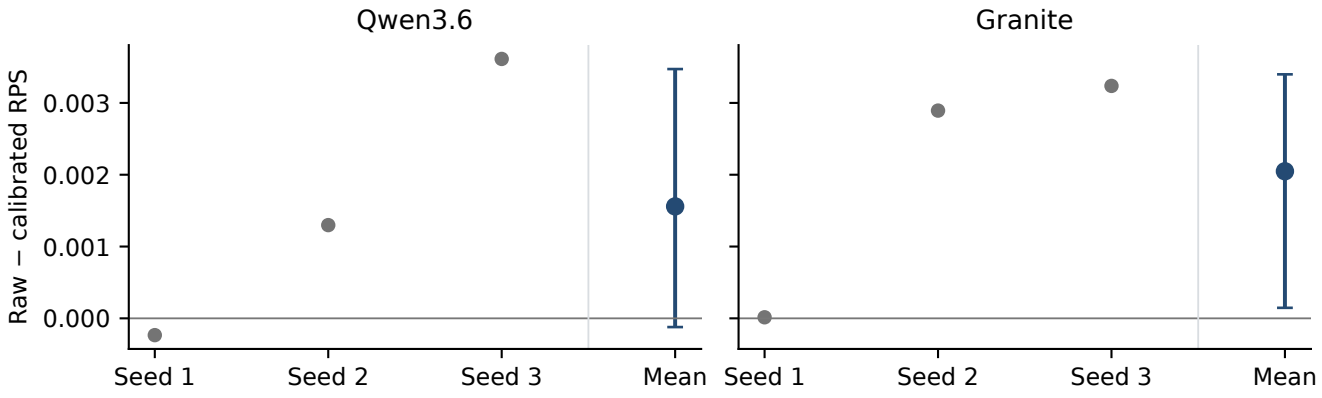



**Figure S2.** All paired MAP-Elites seed effects and overall crossed interval. Seeds are paired by initialization index, not a shared mutator-sample trajectory. Points show independent search runs. Positive favors calibrated fitness.

# 8 Calibration-only readout repeatability

An engineering replay used the same neutral persona, pinned source and serving configuration, and all 120 calibration people twice per model. No search or new evaluation-person forecast was run. A pointwise probability tolerance of $5 \times 10^{-4}$ failed; archived original runs also fail that tolerance against their shared reference. Before reading the new full-panel replay results, a dated amendment fixed a separate calibration-score compatibility criterion: four paired respondent-bootstrap 98.75% intervals, each wholly within $[-0.0001, +0.0001]$ RPS (20,000 draws; Bonferroni-four interval levels). The margin is below 8% of the smaller exploratory GA objective effect. This is a post-failure engineering amendment, not an original scientific preregistration or a claim of population equivalence.

Source, model, configuration, current token reconstruction, repeated token sequences, person/item order, probabilities and costs passed their checks. Three raw-RPS comparisons passed the amended interval rule. Qwen repeat 2 had mean difference $+0.0000700$ and interval $[-0.0001570, +0.0002709]$, which failed. All four comparisons and failed pointwise probes were retained; the planned new-seed search was not released. The observed score variation does not establish its cause, invalidate every original effect estimate, or bound optimized-candidate variability. Historical token sequences were not saved, so present reconstruction cannot retrospectively prove their identity.

Table S4: Calibration replay versus historical neutral readout. Intervals are 98.75% paired respondent intervals; the fixed compatibility margin is ±0.0001. These differences are not optimized-template uncertainty bounds.

| Replay | Mean raw RPS change | Interval | Within margin |
|---|---|---|---|
| Qwen 1 | +0.0000121 | [-0.0000012,+0.0000364] | Yes |
| Qwen 2 | +0.0000700 | [-0.0001570,+0.0002709] | No |
| Granite 1 | +0.0000011 | [0,+0.0000034] | Yes |
| Granite 2 | +0.0000001 | [-0.0000036,+0.0000030] | Yes |

For scale, the saved three-run GA raw-minus-calibrated effects are 0.00163 (Qwen) and 0.00130 (Granite); the planned Granite MAP-Elites effect is 0.00205. The replay intervals concern neutral calibration forecasts, whereas these scientific effects concern selected-template evaluation forecasts. The saved forecast scores reproduce exactly; serving-level repeatability remains open.

**Concurrency intervention.** A subsequent frozen engineering protocol restarted the Qwen server for four blocks with request concurrency $4, 1, 1, 4$, each replaying the same 120-person neutral calibration panel twice. All 57,600 forecasts completed with identical prompt-token hashes. Within-server respondent RPS root-mean-square differences were 0.000118, 0.000979, 0.001071, and 0.000084, respectively. Serial requests therefore did not repair repeatability in this diagnostic. Six of eight historical-reference intervals extended beyond the original ±0.0001 margin, including both replays of the final concurrency-four block despite its small within-server difference. These intervals are descriptive diagnostics, not a new multiplicity-controlled acceptance test. Two blocks per setting do not identify a particular

numerical kernel as the cause or bound optimized-template variation. The additional-seed scientific campaign remains blocked; no failed replay was replaced or tolerance relaxed.

# 9 Selected-template serving robustness

A separate protocol fixed two complete replay rounds before new inference. Each round covers both models, both GA fitness objectives and all three historical template seeds: 24 cells in total, each with a fresh server and the unchanged final runner. Selected templates, model locks, input panels, source hashes, the 8,000-person prior and historical results were preserved. Each cell repeats base/evolved calibration on 120 people and base/evolved/wrong-profile prediction on the same exposed 600 people. There is no new search, prospective respondent sample or hybrid-mixture test.

The primary analysis refits calibration on the original calibration panel. The secondary analysis transfers each historical calibration weight and CDF center to the new evaluation forecasts. Each regime has four model-by-round contrasts and a separate Holm-four correction. The frozen positive screen requires all four positive differences, positive lower crossed-interval bounds and adjusted $p < .05$. Intervals use 5,000 crossed respondent/template draws; sign flips condition on the same three templates. Neither two serving rounds nor resampling those templates estimates variation across new optimization runs.

Table S5: Complete frozen-template serving replay. Positive raw-minus-calibrated GA RPS favors calibrated fitness. Crossed 95% intervals resample respondents and the three fixed templates; respondent sign-flip $p$ values are Holm-adjusted separately across four comparisons per calibration regime. These are two serving rounds on the same exposed people, not new optimizer runs.

| Calibration | Model | Round | Difference | 95% interval | Holm $p$ |
|---|---|---|---|---|---|
| Refitted | Qwen | 1 | +0.00161 | [+0.00109,+0.00215] | 0.0008 |
| Refitted | Qwen | 2 | +0.00159 | [+0.00108,+0.00212] | 0.0008 |
| Refitted | Granite | 1 | +0.00130 | [+0.00059,+0.00203] | 0.0008 |
| Refitted | Granite | 2 | +0.00130 | [+0.00059,+0.00203] | 0.0008 |
| Fixed | Qwen | 1 | +0.00162 | [+0.00109,+0.00215] | 0.0008 |
| Fixed | Qwen | 2 | +0.00160 | [+0.00109,+0.00213] | 0.0008 |
| Fixed | Granite | 1 | +0.00130 | [+0.00059,+0.00203] | 0.0008 |
| Fixed | Granite | 2 | +0.00130 | [+0.00059,+0.00203] | 0.0008 |

Both regimes satisfy their four-contrast screen. All 12 seed-specific primary differences are positive; the complete seed values and cell-level controls are retained in the aggregate report. Calibration weights do not change in any of the 48 base/evolved comparisons with historical values, although CDF centers and probabilities can change. Across the 24 base/evolved evaluations per model, mean refitted-RPS changes versus history range from −0.0000493 to +0.0000783 for Qwen and −0.00000281 to +0.00000109 for Granite. Corresponding raw-RPS changes range from −0.0000851 to +0.0001075 and from −0.00000352 to +0.00000581, respectively. The largest individual probability-coordinate drift is 0.4337 for Qwen and 0.0617 for Granite; preserved aggregate contrasts therefore do not establish pointwise reproduction. Wrong-profile minus evolved RPS remains positive in every cell (Qwen 0.04078–0.04810; Granite 0.03164–0.03782). Ridge fits and their saved losses are unchanged by this experiment.

All 24 cells complete with 2,937,600 confirmed forecasts, zero failed or uncertain forecasts, and 3,596,132,160 reported prompt tokens. Slurm records 29,223 allocated single-GPU seconds (8.1175 GPU-hours), including server startup. This is replay cost, not an end-to-end optimization-efficiency estimate. All four primary contrasts were additionally recomputed from saved individual losses by an independent statistics implementation; its receipt reports the numerical agreement. The historical failed engineering gates remain failed. These results support the observed GA contrast under the two tested serving replays, not a repaired serving configuration or independent confirmation on new people, optimizer trajectories, or behavioral outcomes.

# 10 Additional-respondent fixed-template transfer

After the original outcomes were inspected, a separate internal protocol froze 24 historical selected GA/MAP-Elites-style templates: both models, two search fitness objectives and three old seeds. The new 600 complete-case respondents were sampled from 36,119 eligible people after exclusion of the reachable inventoried historical exposure union. They come from the same public IPIP120 corpus, so lifetime or model-pretraining exposure cannot be excluded. Original Train8000 priors, Calibrate120 alpha and CDF centers, frozen prompts, model snapshots and the selected same-history

ridge specification were reused. No new search or shortlist selection occurred; this test does not evaluate the separate developmental aligned-selector recipe. The internal freeze is not an external preregistration.

The three declared primary tests are Granite calibrated MAP versus raw MAP, calibrated MAP versus its neutral base, and calibrated GA versus its neutral base. Positive differences below are comparator minus calibrated-template RPS. Crossed intervals resample respondents and the three *observed* search seeds; sign flips condition on those seeds. The original combined decision rule requires a positive crossed interval and Holm $p < .05$ within the three-test family. Qwen and other comparisons are diagnostics with their uncorrected sign-flip $p$ values shown for completeness; their small values are not additional declared primary successes.

Table S6: All eight E1 objective and neutral comparisons on the 600 additional respondents. Positive favors the calibrated historical winner. Primary Holm $p$ applies only to the three Granite tests marked P; the Granite MAP calibrated-versus-raw interval crosses zero and fails the combined rule.

| Model | Arm | Contrast | Δ RPS [crossed 95% CI] | Sign-flip $p$ | Primary $p_{\mathrm{H}}$ |
|---|---|---|---|---|---|
| Qwen | GA | cal vs raw | +0.001805 [+0.001358, +0.002257] | .0002 | – |
| Qwen | GA | cal vs base | +0.000337 [−0.000174, +0.000854] | .1184 | – |
| Qwen | MAP | cal vs raw | +0.001680 [+0.000007, +0.003393] | .0002 | – |
| Qwen | MAP | cal vs base | +0.000180 [−0.000506, +0.000806] | .4043 | – |
| Granite | GA | cal vs raw | +0.001229 [+0.000626, +0.001812] | .0002 | – |
| Granite | GA | cal vs base (P) | +0.002286 [+0.001661, +0.002893] | .0002 | .0006 |
| Granite | MAP | cal vs raw (P) | +0.001523 [−0.000182, +0.002733] | .0002 | .0006 |
| Granite | MAP | cal vs base (P) | +0.002156 [+0.001344, +0.003113] | .0002 | .0006 |

Table S7: Mandatory E1 controls for each calibrated historical winner. Values are comparator minus calibrated-template RPS with crossed 95% intervals; negative ridge differences mean ridge is better. These diagnostics were not in the three-test primary family.

| Model | Arm | Same-history ridge minus template | Wrong-persona minus matched template |
|---|---|---|---|
| Qwen | GA | −0.020850 [−0.021898, −0.019877] | +0.044778 [+0.041452, +0.048465] |
| Qwen | MAP | −0.020995 [−0.022161, −0.019898] | +0.041762 [+0.039263, +0.044285] |
| Granite | GA | −0.025188 [−0.026441, −0.023977] | +0.035653 [+0.032601, +0.038636] |
| Granite | MAP | −0.025318 [−0.026688, −0.023941] | +0.035213 [+0.031855, +0.038961] |

All 24 cells have 600 people and 60 target items. The independent output gate confirmed the original frozen lineage, 24 immutable attempts, serving logs, probability and mass arrays, and 2,937,600 requested and confirmed forecasts with no failed or uncertain request. Prompt-token receipts sum to 3,500,088,480; these are new inference costs, not search costs. The separately written finite CPU verifier reconstructed all 72 labelled cell/control score rows, the selected ridge, all 16 contrast distributions and the three-test Holm correction from saved raw forecasts and target labels; its maximum discrepancy from the frozen analysis was $8.33 \times 10^{-17}$. It shares the pinned CDF projection and ridge-fit numerical primitives and does not independently regenerate model forecasts.

The same-history ridge, selected originally and refit on 8,000 old training people with the same 60 observed answers, scores 0.10821 RPS here. Calibrated GA/MAP means are 0.12906/0.12920 on Qwen and 0.13340/0.13353 on Granite. The three observed Granite MAP seed effects against raw are $-0.000288, +0.002381, +0.002477$; the first is unfavorable. A low answer-token mass diagnostic for raw Granite MAP seed 3 is retained (mean 0.9473; 2.37% of target cells below 0.5), above the frozen mean-mass exclusion threshold. Neither favorable base comparisons nor matched-versus-wrong-persona signals establish optimizer superiority to ridge, new-search generalization, or behavioral transfer.

# 11 Exploratory same-shortlist selector intervention

This E2 experiment reused the *original exposed* Evaluate600, rather than E1's additional 600 people. For each of the 12 archived GA trajectories (two models, raw/calibrated search objectives, three seeds), the same five historical candidates received new Select120 forecasts. Raw RPS and a fixed two-fold cross-fitted calibrated RPS each selected one candidate from that identical pool. Both choices were frozen before fresh Calibrate120/Evaluate600 inference, with a fresh neutral-base readback; one coincident pair was deduplicated. There was no new search or new respondent cohort. The selection stage confirmed 432,000 forecasts and the final stage 1,512,000, with zero failed or uncertain requests; prompt-token receipts were 624,369,600 and 1,843,379,400, respectively. These are additional serving costs, not total optimizer costs.

In Table S8, selector effects subtract raw-selector RPS from calibrated-selector RPS within the named search objective. Search effects subtract raw-search RPS from calibrated-search RPS within the named selector. Negative

values favor the first, calibrated option. The interaction is the calibrated-selector search effect minus the raw-selector search effect. Crossed 95% intervals resample respondents and the three *observed* search seeds; respondent sign flips condition on those seeds, with Holm over all ten two-sided tests. A favorable claim requires both a crossed interval excluding zero and adjusted $p < .05$. The three seed columns are differences on the fixed, paired archived trajectories, not new-search replicates.

Table S8: All ten E2 contrasts on the exposed original Evaluate600. Values are calibrated RPS differences as defined in text; lower is better. All four direct selector intervals include zero. The Qwen search effect under the raw selector is the only interval wholly below zero; it is not selector benefit.

| Model | Contrast | Δ [crossed 95% CI] | Seed 1 | Seed 2 | Seed 3 | $p_H$ |
|---|---|---|---|---|---|---|
| Qwen | Selector; calibrated search | −0.000153 [−0.000495, +0.000181] | +0.000044 | −0.000434 | −0.000068 | .3179 |
| Qwen | Selector; raw search | −0.000587 [−0.001456, +0.000044] | +0.000000 | −0.001541 | −0.000221 | .0020 |
| Qwen | Search; calibrated selector | −0.000740 [−0.001712, +0.000308] | −0.001699 | +0.000376 | −0.000896 | .0020 |
| Qwen | Search; raw selector | −0.001174 [−0.001852, −0.000562] | −0.001743 | −0.000730 | −0.001049 | .0020 |
| Qwen | Search × selector | +0.000434 [−0.000152, +0.001131] | +0.000044 | +0.001107 | +0.000153 | .0144 |
| Granite | Selector; calibrated search | −0.000013 [−0.000208, +0.000302] | +0.000249 | −0.000134 | −0.000154 | .8352 |
| Granite | Selector; raw search | −0.000473 [−0.001334, +0.000104] | +0.000062 | −0.001435 | −0.000045 | .0020 |
| Granite | Search; calibrated selector | −0.000171 [−0.001693, +0.001081] | −0.001782 | +0.001124 | +0.000143 | .3179 |
| Granite | Search; raw selector | −0.000631 [−0.001872, +0.000357] | −0.001969 | −0.000177 | +0.000252 | .0020 |
| Granite | Search × selector | +0.000460 [−0.000158, +0.001250] | +0.000187 | +0.001301 | −0.000109 | .0020 |

Calibrated selection changed 11 of 12 winners but none of its four direct effects passes the combined interval-and-$p$ rule. The two raw-search selector tests have small adjusted sign-flip probabilities despite crossed intervals spanning zero; they are not declared improvements. Original selection-audit attempt failed on a post-receipt server shutdown diagnostic. Versioned selection gate classified that event explicitly and passed the archived arrays; final gate and independent CPU readback passed all 12 cells and ten contrasts. This operational chain does not repair numerical serving compatibility. Compared with the historical base, fresh Qwen base per-cell mean absolute RPS-loss drift was roughly 0.00031–0.00037 (maximum individual drift 0.0028–0.0048); Granite drift was much smaller. All reported E2 contrasts therefore compare predictions within the fresh serving execution. E2 does not test its selected winners on E1's additional people, and E1 never evaluated those winners.

## 12 Third-panel Granite-80 frozen-mixture transfer

This E3 test was designed after inspection of the original, E1, and E2 outcomes. Its internal protocol fixed the old Granite calibrated-GA n=80 recipe, controls, four contrasts, 0.001-RPS margin and inference before opening the third panel's target answers; it is not an external preregistration. The 600 complete-case respondents were sampled from 35,519 eligible cases after exclusion of 374,649 reachable inventoried case IDs, including every original split and E1's additional Eval600. They are new to these project panels, not a new sample recruited outside the public corpus; unknown lifetime and model-pretraining exposure remain possible. The old selected prompts, neutral prompts, Granite model snapshot, first-Cal60 scale and CDF centers, old 60-person mixture-selection weights, penalties, categorical/raw features, three disjoint 80-person ridge-fitting subsets, and 8,000-person population prior were reused unchanged. Ridge penalties were chosen on old Select120. The persona search consumed Search80 and Select120 targets, and each new respondent supplies 60 observed answers; "80-person" names only the paired ridge-fit budget. The third-panel labels selected none of these choices.

For each representation the optimized persona plus ordinary ridge was compared with neutral persona plus prior-anchored ridge and neutral persona plus the *same* ordinary ridge. Each neutral arm retained its own old selected mixture weight. The first control changes prompt, weight and ridge centering; the second changes prompt and weight while holding ridge coefficients fixed. Neither is an isolated prompt-only causal contrast. The four planned comparator-minus-hybrid RPS gains form one two-sided Holm family. Before paired respondent resampling, nine fixed per-person loss vectors (three old search seeds by three fixed subset seeds) were averaged, without ensembling forecasts or treating the nine combinations as independent respondents. The fixed success rule requires each gain to be at least 0.001, its paired respondent-bootstrap lower 95% bound to exceed zero, and its Holm-adjusted respondent sign-flip $p < .05$. Both controls must pass for a representation; uncertainty conditions on those old choices. The three n=80 subset seeds have pairwise overlap 0/0/0 and union 240 within Train8000; the same sets are reused across representations and LLM seeds. This index fact does not turn the nine fitted combinations into independent respondent panels or new search replications.

Table S9: All four declared E3 contrasts on the third 600-person panel. Positive gains favor the old optimized mixture; all pass the fixed gate. The two-sided Holm family contains exactly these four rows.

| Features | Neutral comparator | Gain | Paired 95% CI | $p_{\mathrm{H}}$ |
|---|---|---|---|---|
| Categorical | Prior-anchored | .001465 | [.001213,.001722] | .0008 |
| Categorical | Same ordinary | .001843 | [.001595,.002088] | .0008 |
| Raw | Prior-anchored | .001251 | [.001016,.001490] | .0008 |
| Raw | Same ordinary | .001747 | [.001519,.001974] | .0008 |

Table S10: E3 mean RPS component and mixture controls, lower is better. Each row averages the nine fixed per-person loss vectors. The common full 8,000-person same-history ridge scores 0.10995, better than both hybrids.

| Predictor | Categorical | Raw |
|---|---|---|
| Optimized mixture | .132355 | .131532 |
| Neutral + prior-anchored ridge | .133821 | .132784 |
| Neutral + same ordinary ridge | .134198 | .133279 |
| Optimized persona alone | .134180 | .134180 |
| Neutral persona alone | .136691 | .136691 |
| Prior-anchored ridge alone | .138871 | .137463 |
| Ordinary 80-person ridge alone | .140831 | .139406 |
| Full 8,000-person ridge | .109952 | .109952 |

Table S11: Every E3 old-search-seed/subset combination. Subset indices 1–3 denote seeds 202610081–083. $w_O, w_N, w_A$ are the frozen optimized, same-ordinary neutral and anchored-neutral persona weights; $H, N, A$ are their respective third-panel mean RPS values. The nine rows per representation are descriptive fixed combinations, not independent replications.

| Features | Search/subset | $w_O$ | $w_N$ | $w_A$ | $H$ | $N$ | $A$ |
|---|---|---|---|---|---|---|---|
| Categorical | 1/1 | .60 | .55 | .55 | .13229 | .13421 | .13373 |
| Categorical | 1/2 | .65 | .60 | .50 | .13227 | .13415 | .13389 |
| Categorical | 1/3 | .60 | .55 | .50 | .13228 | .13423 | .13384 |
| Categorical | 2/1 | .65 | .55 | .55 | .13235 | .13421 | .13373 |
| Categorical | 2/2 | .70 | .60 | .50 | .13236 | .13415 | .13389 |
| Categorical | 2/3 | .65 | .55 | .50 | .13239 | .13423 | .13384 |
| Categorical | 3/1 | .60 | .55 | .55 | .13239 | .13421 | .13373 |
| Categorical | 3/2 | .65 | .60 | .50 | .13240 | .13415 | .13389 |
| Categorical | 3/3 | .60 | .55 | .50 | .13246 | .13423 | .13384 |
| Raw | 1/1 | .60 | .50 | .50 | .13158 | .13356 | .13295 |
| Raw | 1/2 | .60 | .50 | .45 | .13137 | .13318 | .13275 |
| Raw | 1/3 | .60 | .50 | .45 | .13125 | .13310 | .13265 |
| Raw | 2/1 | .60 | .50 | .50 | .13178 | .13356 | .13295 |
| Raw | 2/2 | .60 | .50 | .45 | .13156 | .13318 | .13275 |
| Raw | 2/3 | .60 | .50 | .45 | .13146 | .13310 | .13265 |
| Raw | 3/1 | .55 | .50 | .50 | .13179 | .13356 | .13295 |
| Raw | 3/2 | .55 | .50 | .45 | .13154 | .13318 | .13275 |
| Raw | 3/3 | .55 | .50 | .45 | .13146 | .13310 | .13265 |

The frozen E3 preparation receipt has SHA256 `76f8a7b57f8b86793697824476105607 3930981c37d750a0b60b4673b9a6bf32`; the new panel's canonical hash is `415583f53de73b7f0295f2c8a289cad1 bab311d9f0ce033a21ba840d6e1f8671`. Preparation reproduced the old recipe and old-Eval losses before serving. The independent output gate passed three immutable one-GPU attempts, each with 72,000 requested/confirmed item forecasts and zero failed or uncertain requests. The total was 216,000 forecasts and 237,089,520 reported prompt tokens, excluding historical search, old calibration and CPU fitting cost. The three allocated single-GPU elapsed times were 694, 622 and 640 seconds (0.5433 GPU-hours total). Seed 3 recorded a post-receipt shutdown diagnostic; its forecast arrays and counts passed the versioned gate, so this is not silently removed. The independently written finite-CPU numerical verifier recomputed all four gains, intervals, Holm values, component means and the full-ridge control from archived raw forecasts and labels. Its maximum absolute discrepancy was $9.38 \times 10^{-11}$, within the protocol's fixed $10^{-10}$ tolerance. This readback did not regenerate model forecasts. Neither this operational pass nor the favorable contrasts validate new search trajectories, the E2 selector winners, Qwen serving compatibility, external behavioral transfer, or superiority to the full ridge.

## 13 Post hoc fixed-weight E3 readback

E4 reuses the already exposed E3 Eval600, frozen Granite prompts, saved probabilities and n=80 ordinary ridge coefficients. Its question and analysis were fixed internally only *after* E3 effects had been inspected; no new respondent panel or model inference was used. At the primary fixed weight $w = .5$, each arm keeps its own historical first-Cal60 scale and CDF center, but evolved and neutral forecasts share exactly the same ordinary ridge and numeric weight. This removes unequal mixture weights from the comparison. It identifies the effect of substituting an *arm-specific calibrated forecast package*, not prompt wording alone. The old last-Cal60 weight-selection labels are part of E3's lineage but their selected weights are unused by this fixed-half primary comparison.

For each representation we average nine fixed *per-person losses* (three historical search seeds by three fixed ridge subsets) before pairing the 600 respondents. The two neutral-minus-evolved fixed-half effects form a separate, two-sided Holm family. The internally fixed descriptive gate requires gain at least 0.001 RPS, positive paired respondent-bootstrap lower 95% bound and adjusted respondent sign-flip $p < .05$. With 5,000 draws and seed 20261021, both pass (Table S12). Intervals condition on these old seeds/subsets and the exposed panel. The numerical verifier recomputed every saved loss and statistic from the raw E3 forecasts with maximum discrepancy zero; it shares the frozen CDF/RPS primitives and does not regenerate model forecasts or create prospective confirmation.

Table S12: E4 fixed-half calibrated-package comparisons on exposed E3 respondents ($N = 600$). Positive is neutral minus evolved RPS, lower scores being better. Both rows form the Holm-2 family and pass the descriptive 0.001-RPS/positive-CI/adjusted-$p$ rule.

| Features | Evolved | Neutral | Gain [paired 95% CI] | $p_{\mathrm{H}}$ |
|---|---|---|---|---|
| Categorical | .132912 | .134389 | .001477 [.001291,.001676] | .0004 |
| Raw | .131831 | .133279 | .001448 [.001264,.001643] | .0004 |

Table S13: Complete *fixed*, unselected E4 weight grid. Each entry is the mean of nine fixed per-person loss vectors. Weight zero reproduces the same ordinary ridge exactly for both arms. Other weights are descriptive; no best weight was chosen from E3 labels.

| Features | $w$ | Evolved RPS | Neutral RPS | Neutral − evolved |
|---|---|---|---|---|
| Categorical | 0 | .140831 | .140831 | .000000 |
| Categorical | .25 | .135723 | .136517 | .000794 |
| Categorical | .50 | .132912 | .134389 | .001477 |
| Categorical | .75 | .132398 | .134447 | .002049 |
| Categorical | 1 | .134180 | .136691 | .002512 |
| Raw | 0 | .139406 | .139406 | .000000 |
| Raw | .25 | .134378 | .135150 | .000772 |
| Raw | .50 | .131831 | .133279 | .001448 |
| Raw | .75 | .131765 | .133793 | .002028 |
| Raw | 1 | .134180 | .136691 | .002512 |

Table S14: Descriptive fixed-half sensitivity to the three *old* Granite search seeds. "Leave out" averages the remaining two seed means, each already averaged across the three fixed n=80 ridge subsets. All values are neutral-minus-evolved RPS; these are not new optimizer replications.

| Features | Seed 1 | Seed 2 | Seed 3 | Leave out 1 | Leave out 2 | Leave out 3 |
|---|---|---|---|---|---|---|
| Categorical | .001579 | .001317 | .001535 | .001426 | .001557 | .001448 |
| Raw | .001546 | .001295 | .001504 | .001399 | .001525 | .001420 |

At the same $w = .5$ but with raw *uncalibrated* persona CDFs, the descriptive neutral-minus-evolved gain is .005634 categorical (evolved .141848, neutral .147482) and .005579 raw (evolved .140851, neutral .146430). These gains are larger than the primary calibrated-package differences, yet both arms' absolute RPS values are worse than their calibrated half-weight counterparts. This secondary comparison has no additional claim threshold or corrected-test status. It neither proves that calibration caused a particular prompt effect nor isolates prompt text from other arm-specific forecast differences.

The three E3 ridge subsets are reconstructed from the exact old 80-of-8,000 seeded permutations; pairwise intersections are 0/0/0 and their union is 240 people. They sit *inside* Train8000, are reused for both representations and all three prompt seeds, and add no unique labelled people beyond Train8000. Before evaluation, the original disjoint Train8000, Search80, Select120 and Calibrate120 roles comprise 8,320 people and 499,200 target-item labels (8,320 × 60). Search80 supplied prompt fitness, Select120 shortlist and ridge-penalty selection, Train8000 supplied the population target prior and full-ridge reference, and Calibrate120 was split 60/60 for E3 calibration and historical weights. Each E3 respondent then supplied 60 observed input answers and 60 separate evaluation targets. E1 and E3 each added 600 inventoried-disjoint same-corpus evaluation people; E4 adds none. These role and item counts are not an equal-supervision comparison of the persona and its n=80 ridge component.

The selected three historical Granite calibrated-GA cells used by E3 account for 1,627,200 successful forecasts, 2,158,592,520 reported prompt tokens and 9.5253 allocated GPU-hours. They are *included in*, rather than added to, the 84-cell original campaign total in Table S15. E3 separately added 216,000 forecasts, 237,089,520 prompt tokens and 0.5433 allocated GPU-hours; E4 added zero GPU forecasts or prompt tokens and used finite CPU jobs. Its freeze, analysis and verifier allocated 96 CPU-seconds in total; the separate historical cost join allocated 8 CPU-seconds. These are allocated core-time, not measured CPU utilization. The independent E4 receipt binds analysis SHA256 `b2d6b3950a6899ba19d535046179eef84dd2471f1fa36b4fef9808bbf61feb71`, the saved individual-loss archive, both primary effects, the fixed grid, and the inherited E3 costs.

Table S15: Source-bound original 84-cell historical cost. Forecast and prompt counts cover successful receipts; GPU time counts task allocations, including failed original search tasks. The five linked smoke/probe jobs have measured allocation but incomplete inference/token accounting. The selected three Granite cells are already inside the 84-cell rows.

| Scope | Confirmed forecasts | Prompt tokens | GPU-h |
|---|---|---|---|
| Search | 32,237,760 | 43,017,723,760 | 215.856 |
| Selection | 3,024,000 | 4,158,424,800 | included |
| Scoped repair allocation | included | included | 54.839 |
| Final | 10,281,600 | 11,979,632,160 | 28.729 |
| 84-cell array total | 45,543,360 | 59,155,780,720 | 299.424 |
| Linked smoke/probe jobs | unavailable | unavailable | 1.176 |
| Measured historical minimum | at least 45,543,360 | at least 59,155,780,720 | 300.599 |

The 84-cell join required all 84 original search, 12 repair and 84 final Slurm task rows. Search had 72 completed, two failed and ten cancelled task rows; repair and final completed. GPU-hours sum allocated GPUs times elapsed seconds, including startup, and exclude queue time and duplicate batch rows. Earlier smoke jobs 8692 and 8808 and context probe 8918 contribute another 4,233 GPU-seconds, linked by contemporaneous queue/recovery records. This 300.599-hour minimum predates and is outside the current-cycle 60 GPU-hour planning cap. Other historical replays, failed/cancelled request work, mutator/completion tokens, CPU work and dollars are unresolved. Successful receipt totals are not all incurred serving work, and no end-to-end efficiency ranking follows.

## 14 Unexecuted E5 replication and old-only engineering

A planned replication with three new Granite optimizer seeds and a disjoint 600-person same-corpus panel (E5) remains unexecuted: its strict readout gate failed before search release. No E5 optimizer search, new-panel forecast or holdout-target analysis was run. The engineering checks in Table S16 use the original Cal120 forecasts and runtime metadata only; they add neither a transfer sample nor a new optimizer result. The reference remains the original calibrated-GA Granite seed-1 `base_cal` archive (SHA-256 prefix `9ebeffc8f789`), with the unchanged rule $\max|\Delta p| \le 5 \times 10^{-4}$.

Table S16: Old-only E5 engineering and independent readback. Completion or verification of a diagnostic does not pass historical compatibility or release search. Profile comparisons use their common respondent rows.

| Job | Recorded result |
|---|---|
| 23135 | Two fresh sequential servers, explicit batch-invariant OFF/ON: 30,720 confirmed forecasts and 24,258,640 prompt tokens. OFF fails five of six profile pairs; ON probabilities are byte-identical in all six. Both arms fail all four comparisons with the canonical reference; the maximum probability difference remains 0.0303456. |
| 23637 | Independent CPU readback of archived raw top-20 log probabilities and reconstruction of answer mass and normalized arrays, with checks of prompt tokens, source lineage and costs, agrees with the diagnostic; historical compatibility still fails. |
| 23645 | All 42 historical Granite neutral-Cal arrays, two preserved replays and eight OFF/ON profiles are retained. The canonical exact mode recurs in six historical cells. The 42 historical full-Cal arrays have 27 exact modes but one shared exact tail over respondents 9–120. |
| 23654 | Fixed initialization-to-post-capture overlap across all 42 launches: 0/4 overlapping pairs and 27/857 nonoverlapping pairs share a mode; proportion difference $-0.0315053$, two-sided task-label permutation $p = 1$ (100,000 draws). The fixed overlap proxy has no positive support. |

The invariant ON path stabilizes this execution without recovering the historical reference; it does not identify a single numerical kernel. Repeated canonical bytes weaken a one-off corruption hypothesis without proving absence or cause. The overlap test is observational, has only four exposed pairs, and assumes task-label exchangeability that temporal or launch-order dependence may violate. Its interval definition was not changed after the result. Recorded selected attention backend, AOT action and graph mode are constant across the 42 launches despite the multiple output modes. An older fresh-server full-Cal first call and repeat with identical rendered tokens (job 9149) already both fail the same canonical gate; this is not a new E5 replication. These checks establish no new-search-seed robustness or mechanism-level cause. Full receipt hashes and claim locators are retained in `E5_ENGINEERING_PAPER_BINDING_20261024.json`.

The additional E5 engineering allocations (jobs 23036, 23121 and 23135) consume 0.188333 GPU-hours, increasing the prior current-cycle total from 13.796111 to 13.984444 GPU-hours; job 23135 accounts for 0.094444 GPU-hours of that increment. The independent CPU checks add no GPU allocation. A later mechanism-trace allocation (24116) fails after one second, before server launch or forecasts, adding 0.000278 GPU-hours and bringing the current-cycle total to 13.984722 GPU-hours. Its client-environment preflight validation requests Triton package metadata unavailable in that interpreter; no hidden-state, logit or sampler observation is obtained. That attempt is closed without a retry and supplies no layer-localization evidence. The separately inventoried historical minimum remains 300.599444 GPU-hours (Table S15); it is outside this current-cycle total, and the older job 9149 is outside the current-cycle ledger. Neither total is an end-to-end project cost or an efficiency result.

A subsequent raw-only CPU check (24086) tests the narrower hypothesis of unchanged digit-relative values plus a shared row offset and final round-to-nearest FP32 storage. In the OFF full-Cal repeat contrast, 8 of 28,800 anchored digit components across four forecasts and two respondents exceed the stated component-specific rounding bounds; the maximum change is 0.250000119, versus a maximum bound of $1.91 \times 10^{-6}$. All six within-ON contrasts are exactly zero. This rejects that stated offset-and-final-rounding model, not every possible internal arithmetic error, and identifies neither a kernel nor a canonical repair. The failed JSON publication of its earlier implementation (24083) is retained separately and is not interpreted as a numerical result.

Static opcode inspection, without deserializing or executing the AOT payload, finds 21 embedded source texts in the current cached artifact at the historically recorded path; all match the collected current sources (15 vLLM and six Torch/Transformers files). This is a partial cached-source witness for the compiled forward and support code. No October 7 artifact hash exists, so it does not authenticate the historical bytes or the complete scheduler, attention-kernel and runtime environment. Engineering-only preflight failures (24108, 24110, 24112) and the successful CPU compile check (24113) are retained; none is scientific parity evidence.

A separately authorized corrected trace (24315) validates Triton in the pinned vLLM interpreter and the supervisor in its client interpreter (CPU preflight 24313). One default-OFF server performs exactly two old-Cal120 reads: 14,400 confirmed forecasts and 11,369,760 prompt tokens. The GPU job remains FAILED because its final join expects external request IDs, while vLLM adds an eight-hex-character internal suffix. Its 14,400 captured rows and raw artifacts are retained unchanged. A source-derived, posthoc CPU-only binding correction (24328) requires unique raw and normalized IDs, an exact 14,400-row client/server bijection, recomputed request hashes, and unchanged artifact/source hashes before and after analysis. The two additional source files explaining the internal suffix are current post-run witnesses, absent from the original seven-source runtime receipt. The earlier CPU binding implementation failure (24322, before array analysis) is retained.

The bound trace contains 7,200 paired forecasts. Ten pairs differ already in the exact BF16 LM-head-input hidden-state bits; all ten also differ in returned top-logprob records. This localizes observed repeat variation at or before that hidden-state boundary under instrumentation, without identifying an earlier layer or kernel. All ten have different final sampler-call shapes and repeat-local call ordinals, but these contexts also differ widely among unchanged pairs and do not identify a causal batch effect. The other 7,190 pairs have identical hidden states and five captured canonical-digit logits; their 71,900 anchored digit-pair comparisons show no sampler-repeat variation beyond the stated bound. No isolated head variation is observed with identical hidden states. These nulls cover only selected coordinates and the stated arithmetic model. Snapshot kernels and retained buffers can alter scheduling; this trace does not establish the uninstrumented historical cause, repair canonical compatibility, or release E5 search. The one GPU allocation lasts 134 seconds (0.037222 GPU-hours), bringing the current-cycle total to 14.021944 GPU-hours. No additional GPU replay or new-target inference is performed for the CPU binding correction.

Retained-artifact CPU forensics (24574), prespecified over all 7,200 pairs, finds zero token-signature, token-count or selected-position mismatches; every selected position equals the prompt token count minus one. These are receipt checks: the actual token-ID arrays were not retained. All ten changed hidden-state pairs lie in the previously fixed first-eight-respondent group (10/480 forecasts), versus 0/6,720 in the other 112 respondents. Each changed vector differs numerically in 3,390–3,763 of 4,096 coordinates, with no signed-zero-only changes; the least-squares common-scale residual is 0.7595–1.4154% of the second vector's norm. These descriptive metrics do not identify RMSNorm, KV-cache or batch scheduling as a cause. The tensors correspond to selected final-RMSNorm outputs supplied as LM-head inputs; the snapshots are queued only after the head call returns, and backend nonmutation is not independently proved. Earlier-layer states were not retained, so the earliest upstream divergence remains unobserved.

A CPU-only static AOT feasibility audit (24710; 203 seconds, two CPUs, 8 GiB, no GPU) inspects the unchanged 17,075,197-byte cached object using pickle opcodes and Python ASTs without deserializing or executing payloads. It inventories seven complete and five incomplete pickle units, 185 AST-readable Python-source records (34 proof-eligible), and 63 opaque binary units, including a 162,874-byte execution-code witness. The bounded audit does not prove complete producer–buffer–consumer maps for the 66 boundaries, complete original numeric-launch association, or an exact existing logits-index delivery channel; the runner's dynamic input expansion remains unresolved. The verdict is

FEASIBILITY_HOLD / OBSERVER_UNAVAILABLE, not a proof that another static method could never recover these relationships. The fixed protocol therefore stops before observer implementation or GPU submission, with no eager fallback, signature change or numerical recompile.

# 15 Full-data representation comparison

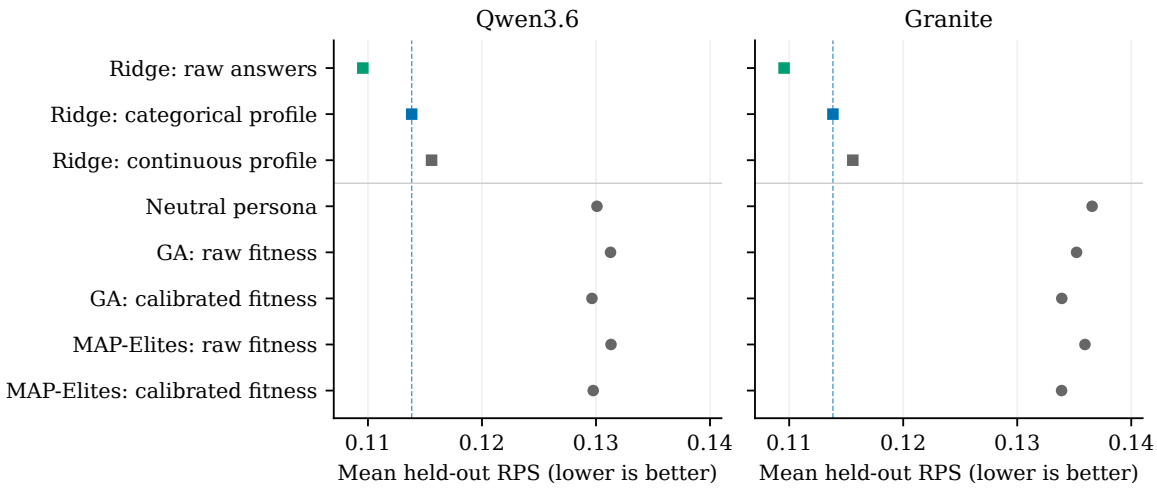



**Figure S3.** Full-data ridge and persona controls. Shared statistical predictors are repeated across panels for comparison. All ridge controls in this figure use 8,000 person-level labelled mappings. LLM arms use their frozen population calibration; ridge directly predicts ordinal CDFs. These are held-out RPS values, not an equal-supervision comparison. The dashed vertical line marks categorical-profile ridge.

# 16 Associative label-budget diagnostics

Two additional 48-test families cover the ordinary and shared-prior budget diagnostics, separately from the earlier 33 representation and nine objective contrasts.

The original 600-person cohort was already exposed. The ordinary-ridge diagnostic was followed by a separate rival test using the 8,000-person empirical item prior. Both are post hoc. For each fitting subset, the anchored variant predicts

$$\widehat{F}_{\text{anchored}}(x) = \text{Proj}\big(\widehat{F}_{\text{ridge}}(x) + F_{0,8000} - \overline{Y}_{\text{subset}}\big), \tag{1}$$

where $\widehat{F}_{\text{ridge}}$ is the unprojected intercept-fit ridge output and $\overline{Y}_{\text{subset}}$ the mean target-threshold vector. This changes only the intercept at a given penalty; each variant selects its penalty on the same 120 selection people. It is algebraically centered no-intercept ridge about the population prior. Using the original solver precision makes the full 8,000-point correction zero and exactly reproduces every original individual loss.

Table S17: Mean RPS at each ridge fitting budget. Lower is better.

| Features | Fit people | Ordinary | Shared prior |
|---|---|---|---|
| categorical | 80 | 0.13956 | 0.13802 |
| categorical | 200 | 0.13150 | 0.13099 |
| categorical | 320 | 0.12791 | 0.12759 |
| categorical | 1,000 | 0.12069 | 0.12059 |
| categorical | 8,000 | 0.11384 | 0.11384 |
| categorical | exact 80 | 0.13814 | 0.13669 |
| raw | 80 | 0.13795 | 0.13643 |
| raw | 200 | 0.12972 | 0.12921 |
| raw | 320 | 0.12607 | 0.12575 |
| raw | 1,000 | 0.11824 | 0.11815 |
| raw | 8,000 | 0.10955 | 0.10955 |
| raw | exact 80 | 0.13624 | 0.13494 |

Positive contrast values in the accompanying diagnostic files mean LLM minus ridge RPS; negative values favour the LLM. Every variant has 48 comparisons: 2 representations times 6 fitting conditions times 4 GA model/objective arms. Each family is separately Holm corrected. This does not correct for adaptively choosing the second diagnostic

after seeing the first. The full 48-row tables, all subset means and frozen selection specifications are retained in the local research artifact; no public data-release claim is made.

The ten fitting subsets per budget overlap. Their mean losses average separate predictors, not ensemble probabilities; their min/max spread is not a confidence interval. Crossed intervals resample only respondents and LLM search seeds, conditioning on statistical fits and the population prior. Fitting counts omit 120 selection people. LLMs additionally use 120 calibration people and foundation-model pretraining, so matching the empirical prior does not imply identical total supervision or prove a causal sample-efficiency advantage. The exact 80 row uses the original search people instead of training-panel subsets.

# 17 Fitting-budget visualization

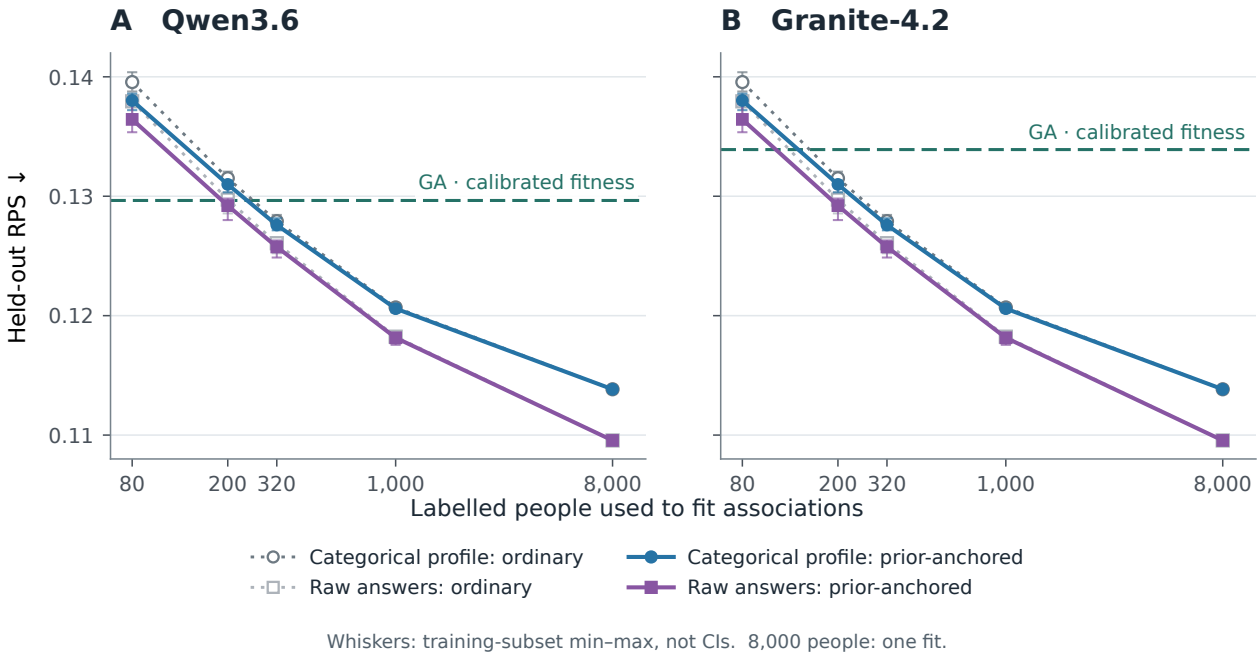



**Figure S4.** Ordinary and shared-prior ridge learning curves. Whiskers are ranges across ten overlapping fitted subsets, not confidence intervals. At 8,000 there is one fit. LLM lines average three frozen runs; statistical curves repeat across model panels for comparison.

# 18 Exploratory complementarity protocol

We apply the established linear forecast pool Gneiting and Ranjan [2013]:

$$H_{ij} = (1 - w)R_{ij} + wP_{ij}, \qquad w \in \{0, .05, \ldots, 1\}. \tag{2}$$

The convex combination is a valid CDF because both components are valid CDFs. Ridge uses categorical or raw-answer features and nested subsets of 80, 200 or 320 training people, defined by seeds 202610081–202610083. It selects its penalty on Select120. The first 60 ordered Calibrate people fit each persona's scale and mean CDF; the last 60 choose the mixture weight. Neither calibration nor mixture weights are refitted after selection. All three calibrated-GA runs and their own neutral counterparts are retained. No new LLM calls are made.

The empirical prior uses Train8000 and the original prompt search uses Search80/Select120. Each mixture requires the same last60 selection labels; neutral and population mixtures independently select their weights. Thus all supervision roles remain explicit. Train subsets overlap Train8000 and do not represent additional distinct people. RPS first averages 60 items per person, then averages losses over nine fixed fit/run combinations. This is not an ensemble of nine prediction vectors. A single 48-test Holm family covers all 12 conditions against four controls. Intervals and sign flips use 5,000 paired respondent draws with seed 20261008, conditional on these fixed fits. A useful screen result requires gain at least 0.001, a positive 95% interval and Holm $p < .05$. All comparisons and nulls are retained. Selection parameters were saved before explicit evaluation extraction; the full corpus and evaluation-bearing

result containers had already been loaded. This is a procedural dataflow guard, not capability-level isolation; original full120-calibrated losses reproduce exactly in all 12 base/evolved checks.

# 19 Complete complementarity results

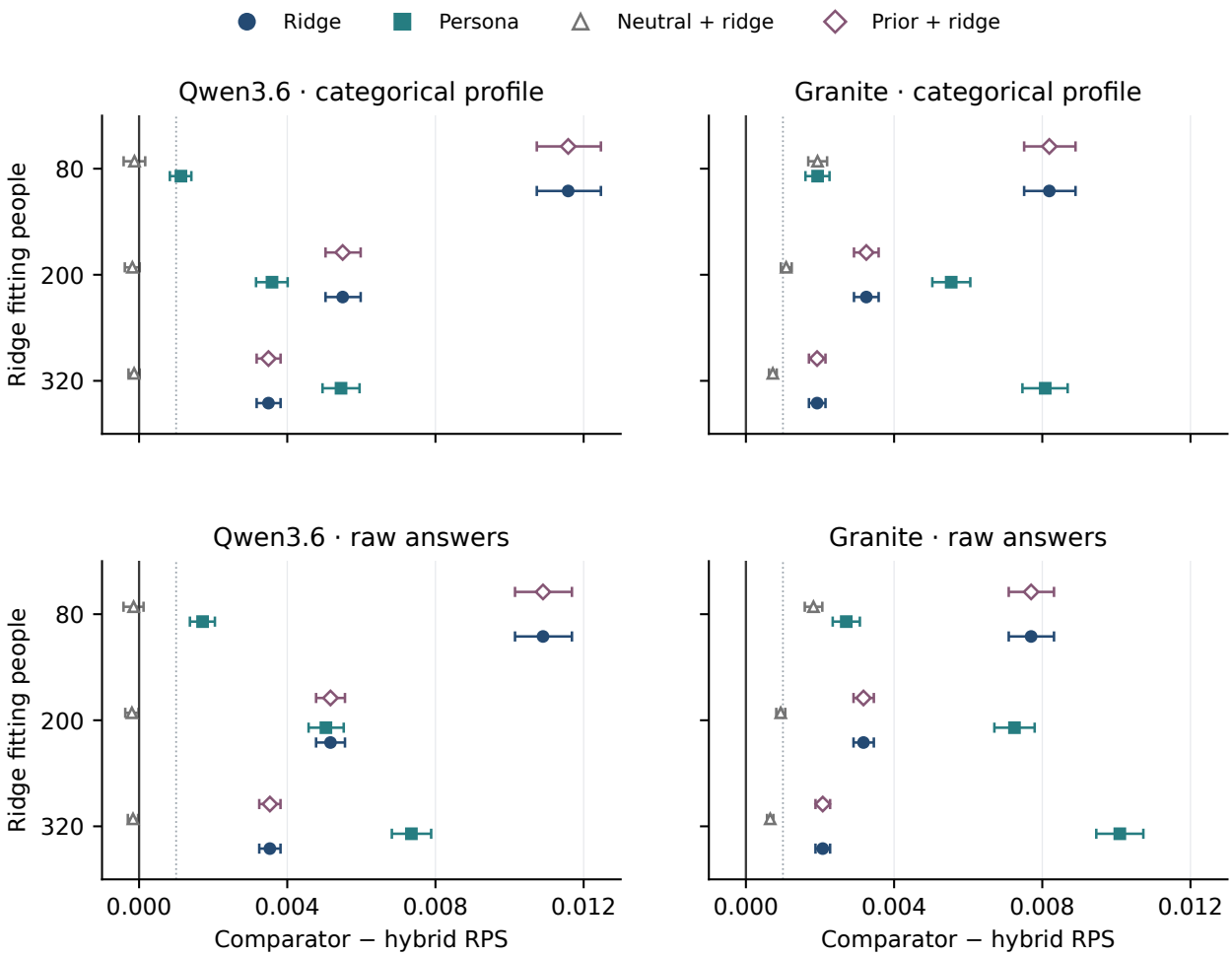



Figure S5: All 48 contrasts: control minus optimized-persona mixture RPS. Positive favors the mixture. Bars are respondent-bootstrap 95% intervals conditional on the fixed nine fits; the dotted line is the 0.001 screening threshold. Color and marker identify the comparator. Full numerical means, intervals and within-family Holm values are retained in the associated table.

Mixtures beat both standalone components in every condition. Optimization itself passes the additional magnitude/statistical gate over the neutral mixture only on Granite for categorical 80/200 and raw 80. No Qwen condition passes that gate. The selected prior-mixture weights are all zero. Prior mixing is a different comparator from refitting the intercept around the population prior; the latter is tested separately below. The analysis is adaptive on exposed respondents and complements, rather than supersedes, earlier corpus-specific null evidence on persona combinations.

## 20 Adversarial prior-anchored comparison

After viewing the original screen, a separate protocol froze all 12 conditions and the original hybrid forecasts. The two new rivals are the prior-anchored ridge defined above, selecting its penalty on Select120, and a neutral-persona mixture using that ridge. Neutral calibration uses the first 60 Calibrate people and its independently selected mixture weight uses the last 60. All 18 anchored fits reproduce the corresponding earlier individual losses within $8.4 \times 10^{-17}$; original hybrid means are unchanged exactly. No new LLM calls or GPU inference are used. The procedural evaluation guard does not restrict the process's underlying file access.

The 24 contrasts form a separate Holm family with 5,000 paired respondent bootstrap/sign-flip draws, conditional on the same nine fixed combinations. The unchanged screening rule requires gain at least 0.001, positive marginal 95% interval and adjusted $p < .05$. All twelve hybrids beat anchored ridge by 0.00161–0.00984 RPS (all $p_H = .0048$). Against neutral-plus-anchored ridge, Granite gains 0.00044–0.00164; only categorical 80 and raw 80 also clear the magnitude threshold. Qwen loses 0.00025–0.00038 in all six conditions; five differences have $p_H < .05$ and categorical 80 has $p_H = .0514$. Thus the stronger rival preserves the component-combination result while narrowing the case for optimized personas. This remains adaptive exposed-cohort evidence. Changing both the prompt and decoder makes the rival a practical comparison, not a causal ablation of optimization.

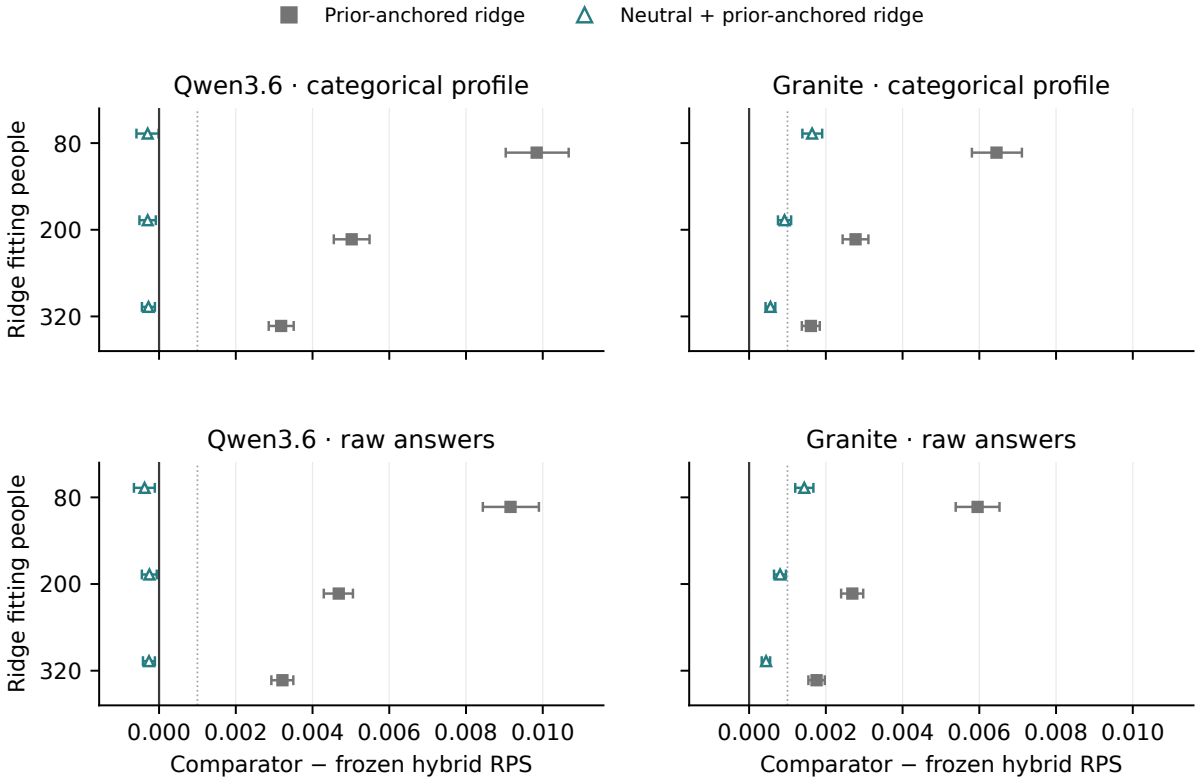



Figure S6: All 24 adversarial contrasts: comparator minus the unchanged original hybrid. Bars are conditional respondent-bootstrap 95% intervals; positive favors the hybrid. The dotted line is the 0.001 screening threshold. All conditions and unfavorable outcomes are retained; this family is separate from the original 48 contrasts.

# 21 Reproducibility specification

**Common execution contract.** Every cell used the same Qwen/Qwen3.8-27B mutator, a strict mutable JSON genome, temperature 0.7, at most 2,500 generated tokens per ordinary mutation, disabled thinking, and a fresh seeded mutator request. The serialized genome has exactly four top-level fields: `role_definition`, fixed-length arrays of five `trait_formulations` and 30 `facet_formulations`, and `critic_formulations`. The runtime genotype also carries fixed intensity modifiers copied from the neutral template; they are never mutated. Malformed proposals consumed no forecast budget, whereas any mutator transport failure invalidated the cell. Search was capped at $B = 80$ full optimization-panel equivalents. Since the panel has 80 respondents and 60 target items, the cap is $80 \times 80 \times 60 = 384{,}000$ respondent–item forecasts per cell; realized search usage ranges from 78.2 to 80.0 equivalents. The separate shortlist stage rescored at most five candidates on 120 respondents (7,200 forecasts per candidate; at most 36,000). It selected the lowest raw RPS with the genotype hash as a deterministic tie break. Thus the `generations`, `rounds`, and `iterations` fields below describe the planned schedule; the forecast ledger, rather than those fields, was the operative stopping rule.

Table S18: Exact initialization, update/history, and budget-relevant parameters for the eleven search methods. "Seed" is the neutral genotype specified below; all fitness comparisons minimize the cell's declared optimization objective.

| Method | Initialization | Update and retained history | Fixed parameters |
|---|---|---|---|
| GA | Seed plus mutations to population 8. | Two elites; two independent tournament-3 parents. LLM crossover with probability .8 when parents differ; a crossed child is mutated with probability .3 and a copied parent is always mutated. | Population 8; elites 2; tournament 3; crossover .8; mutation .3; configured 10 generations. |
| Differential evolution | Same seed-plus-mutations population as GA. | For each member $x$, sample three distinct other members $a, b, c$; the LLM applies the $a-b$ difference to $c$ and crosses the result with $x$; greedily replace $x$ only on improvement. | Population 8; configured 10 generations. The recorded crossover-rate field is .8, but the deployed loop does not use it as a stochastic gate. |
| Prompt-Breeder | Seed, then seven mutations using successive members of a fixed eight-instruction mutation-prompt pool. | One elite; tournament-3 parent; each member carries its mutation instruction. With probability .25 the LLM rewrites that instruction (temperature .9, 200 tokens), then applies it to the parent genotype. | Population 8; instruction pool 8; hypermutation .25; configured 10 generations. |
| MAP-Elites style | Two islands, four initial evaluations per island; only the first is the unchanged seed, all others are seed mutations. | Uniformly sample an occupied archive cell; mutate using the last six lineage records (operation, score, prompt length, predicted SD); replace a cell only on lower score; migrate the island champion every 12 iterations. | 2 islands; $4 \times 4$ bins with length edges $[0, 1500, 2500, 4000, 100000]$ and SD edges $[0, .8, 1, 1.2, 9]$; 4 initial points/island; history 6; configured 72 iterations. |
| Local GEPA | Evaluate seed on the full panel and initialize a respondent-wise Pareto pool. | Select a Pareto member in proportion to its number of respondent-wise wins; compare parent and reflected child on the same eight respondents; give the reflector the ten worst target-item errors; run a full evaluation only after a minibatch improvement. Retain the Pareto front and global best; spend remaining whole equivalents on best-candidate mutations. | Minibatch 8; at most 12 Pareto candidates; 10 feedback items; configured 80 evaluations. |

Table S19: Search specifications, continued. The same executed-forecast budget applies.

| Method | Initialization | Update and retained history | Fixed parameters |
|---|---|---|---|
| Official GEPA | Seed candidate and optimization respondents indexed 0–79. | GEPA v0.1.4 Pareto selection with per-example metric $-\text{RPS}_{raw}$; reflection minibatches of 8; merge disabled; no metric cache. The external forecast ledger is authoritative and returns the last fully scored elite at exhaustion. | Reflection minibatch 8; package v0.1.4. |
| Native GigaEvo | Exact neutral JSON written as the initial program. | Pinned native guided pipeline; single island without distant parents; no memory; fitness $1 - \text{RPS}$; external validator and forecast stopper. | Internal v2.20.1 at commit `f72ab08c`; maximum 160 mutants; concurrency 1; $8 \times 8$ behavior space over length $[0, 16000]$ and predicted SD $[0, 2]$. |
| OPRO | Evaluate seed. | Propose in blocks of 8 from up to the 12 best valid scored genotypes, printed from worse to better within that retained set. | 8 proposals/round; history 12; configured 10 rounds. |
| ProTeGi | Evaluate seed and start a beam. | For every beam member, generate two textual-gradient edits using the ten worst target-item errors; evaluate one edit per gradient; retain the best four candidates. | Beam 4; 2 gradients per candidate; 1 edit/gradient; configured 20 rounds. |
| Random search | Evaluate seed. | Draw $B - 1$ independent mutations of the seed; return the best full-panel candidate. | 80 evaluations. |
| Best-of-$B$ | Evaluate seed. | Draw $B - 1$ independent seed mutations with the initial-population generator at temperature 1.0; return the best full-panel candidate. | 80 evaluations; mutation temperature 1.0. |

**Neutral genotype and exact rendering.** The neutral role is: "You are a simulated respondent. Answer each statement according to the supplied personality profile." For each of the five canonical traits $t \in \{$neuroticism, extraversion, openness, agreeableness, conscientiousness$\}$, its exact formulation is "Tendency toward $t$." For each canonical facet label $f$ in the item key, the exact formulation is "Tendency toward lowercase($f$)." The critic is: "Use only the supplied profile. Interpret the response scale consistently." Scores use boundaries $[0, 20, 40, 60, 80, 100]$ with labels *very little*, *slightly*, *moderately*, *quite strongly*, and *very strongly*; operational intervals are $[0, 20)$, $[20, 40)$, $[40, 60)$, $[60, 80)$, and $[80, 100]$ after endpoint clipping.

For a genotype $(r, T, F, c)$, the exact system-prompt skeleton is `r\nYour profile:\n`, followed first by one line per trait and then one per facet: `- This trait (NAME) describes you MODIFIER: DESCRIPTION` and `- This facet (NAME) describes you MODIFIER: DESCRIPTION`. It ends with `\n\nInternal reflection guideline:\n` and $c$. A facet score is $100(\bar{x} - 1)/4$ from its two observed items; a trait score is the mean of its six facet scores.

**Selected wording example.** The frozen Qwen calibrated-GA seed-1 template changes the neutral "Tendency toward extraversion." to "The intensity with which you derive energy from social interaction, assertiveness, and positive emotional experiences." Its critic includes: "Avoid converging on a central tendency; instead, maximize the semantic variance between answers to accurately reflect the specific magnitude indicated, ensuring that a 'very little' response is distinctly different from a 'slightly' one in both tone and content, so that your agreement or disagreement scales linearly with the provided intensity." These are exact excerpts from one selected template, showing the intervention on profile interpretation. They neither characterize every optimized template nor establish a causal mechanism. The complete genotype accompanies the source notes.

**Items, people, and partitions.** Items are grouped by facet after sorting by item number. The first two items of each of 30 facets form the 60-item observed input; the remaining two form the disjoint 60-item prediction target. Eligible rows have all 120 item values in $\{1, 2, 3, 4, 5\}$, unique case IDs, and no match to the inventoried historical exposure set. Rows are sorted by case ID, permuted with seed 20261004, and sliced without overlap into train/optimization/selection/calibration/evaluation panels of 8,000/80/120/120/600 people (8,920 reserved rows). A subsequent inventory records 410,168 complete cases, of which 374,049 match the accumulated exclusion set and 36,119 remain eligible in that inventory. These are inventory counts, not a claim of new untouched test people; all-host exposure coverage is unproven. The local reproducibility bundle does not include the raw corpus, a verified dataset license, or validated demographic summaries. Sex, age, and country are loadable fields, but no demographic interpretation or distribution is asserted here.

**Frozen exemplar and cryptographic closure.** The saved Qwen calibrated-GA seed-1 selection has raw selection RPS 0.1692038488, five shortlisted candidates, and canonical selected-record hash `4131b9d1dc37cd4f`; its complete runtime genotype is transcribed in the accompanying source notes. A second located exemplar, Granite calibrated MAP-Elites seed 1, has selected-record hash `09d79eb1c951f4f8`. The canonical manifest hash is `9037e61283d70779`; panel hash `786de40844429e19`; frozen-record file hash `155b512315b52679`; final 84-cell recomputation hash `7a3009c03446254d`; model-lock file hash `342d9215e991e973`; item-key hash `2cd1483c7d3d539c`; and deployed source-closure hash `e333de246887ab0c`. Full 64-hex digests, model revision pins, source anchors, and the exact exemplar appear in `REPRO_SOURCE_NOTES.md`.

**Statistical predictors and score translation.** The 60 ordinal targets are encoded as 240 indicators $\mathbf{1}[y_j \le k]$ for $k = 1, \ldots, 4$. Raw inputs use 300 one-hot indicators; categorical inputs use 175. Ridge fits an intercept with the Cholesky solver, without scaling these one-hot features. Select120 minimizes RPS over penalties $\{1, 10, 100, 1000\}$, with smaller penalties breaking ties. Continuous-profile features alone are standardized using training statistics. Predictions undergo the exact squared-error projection of each four-threshold vector onto $0 \le F_1 \le \cdots \le F_4 \le 1$. ExtraTrees uses 64 trees, minimum leaf size selected from $\{10, 30\}$, feature fraction .8, random seed 20261004, one worker, and otherwise scikit-learn defaults. Answer-equivalent curves use a nonincreasing isotonic fit and linear interpolation over the registered budget grid; out-of-grid values remain censored. Their bootstrap resamples respondents and the three fixed input orders, not training fits.

**Pinned model revisions.** The full model revision pins are:

Qwen predictor: `995ad96eacd98c81ed38be0c5b274b04031597b0`
Granite predictor: `9e668ce1c538387ef24d3644e9b0606647762636`
Qwen mutator: `1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0`.

The documented analysis environment uses Python 3.12.14, NumPy 2.3.5, scikit-learn 1.7.2, and serving vLLM 0.28.0. Full source notes and receipts are retained locally; no publicly executable anonymous artifact is claimed.

Table S20: All exploratory mixture conditions: mean RPS, lower is better. Each mean averages losses of the nine fixed fits. The preceding forest plot shows every paired contrast and its conditional interval.

| Model | Features | Fit $n$ | Hybrid | Ridge | Persona | Neutral mix | Prior mix |
|---|---|---|---|---|---|---|---|
| Qwen | categorical | 80 | 0.12859 | 0.14017 | 0.12972 | 0.12846 | 0.14017 |
| Qwen | categorical | 200 | 0.12613 | 0.13163 | 0.12972 | 0.12595 | 0.13163 |
| Qwen | categorical | 320 | 0.12427 | 0.12776 | 0.12972 | 0.12413 | 0.12776 |
| Qwen | raw | 80 | 0.12800 | 0.13891 | 0.12972 | 0.12786 | 0.13891 |
| Qwen | raw | 200 | 0.12468 | 0.12984 | 0.12972 | 0.12448 | 0.12984 |
| Qwen | raw | 320 | 0.12237 | 0.12590 | 0.12972 | 0.12221 | 0.12590 |
| Granite | categorical | 80 | 0.13198 | 0.14017 | 0.13392 | 0.13392 | 0.14017 |
| Granite | categorical | 200 | 0.12838 | 0.13163 | 0.13392 | 0.12946 | 0.13163 |
| Granite | categorical | 320 | 0.12584 | 0.12776 | 0.13392 | 0.12656 | 0.12776 |
| Granite | raw | 80 | 0.13121 | 0.13891 | 0.13392 | 0.13303 | 0.13891 |
| Granite | raw | 200 | 0.12667 | 0.12984 | 0.13392 | 0.12761 | 0.12984 |
| Granite | raw | 320 | 0.12383 | 0.12590 | 0.13392 | 0.12448 | 0.12590 |

## 22 AI-assistance record

This revision additionally used Claude Code for independent manuscript critique and Codex for analysis, editing and plotting; the main paper states the combined disclosure. Historical records below concern earlier stages.

Codex assisted source inspection, experiment specification, coding and writing; retained review records identify GPT-5.6-Sol for draft and claim review. Retained Claude artifacts document generic figure-layout assistance; research labels and numerical values were bound and checked locally. Available records are inventoried in `AI_CONTRIBUTION_LEDGER_20261020.json` in the research evidence directory, including artifact hashes and coverage limits. This is a partial record: complete prompt/response retention and independently attested model identities are unavailable. Exact versions and prompts for older unrecorded interactions remain unknown.

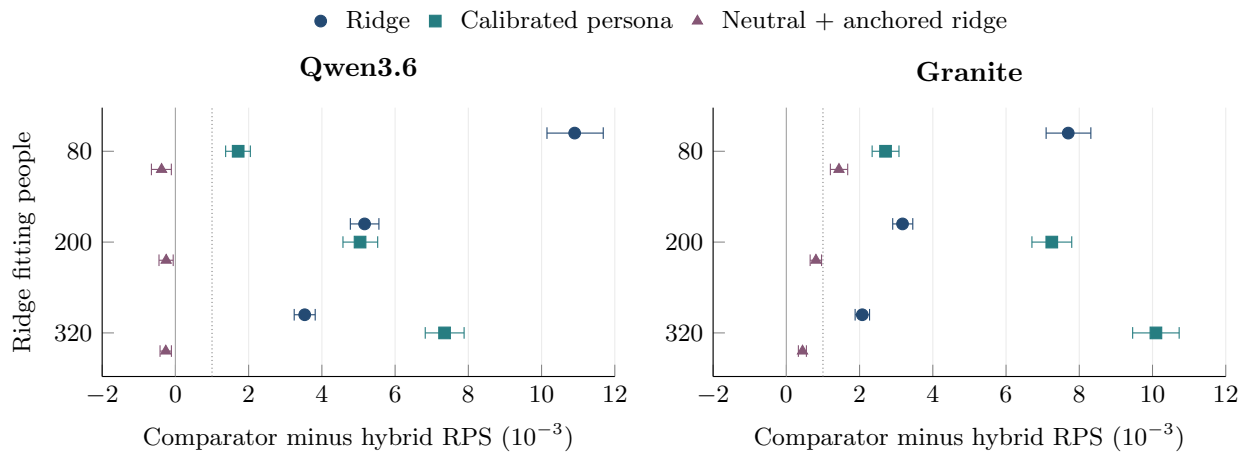



Figure S7: Exploratory paired gains on the exposed 600 respondents (raw-answer features). Positive values favor the optimized-persona mixture; bars are 95% respondent intervals conditional on nine fixed fit/run combinations. The vertical dotted line marks the 0.001 practical threshold. Component comparisons belong to the 48-test family; the stronger neutral rival to the separately frozen 24-test family. Intervals are pointwise. All LLMs also use the 8,000-person prior; small fitting counts are not total supervision budgets. The supplement includes categorical features and every declared comparison.

# 23 Exploratory mixture uncertainty

# 24 Original-cohort objective effects

# 25 Detailed original-cohort budget and mixture diagnostics

## 25.1 Exploratory fitting-budget diagnostics

On the exposed cohort, shared-prior statistical controls have higher mean RPS than calibrated GA at 80 fitting people but lower mean RPS at 320; the supplement reports full learning curves and supervision counts.

Using the exact 80 search respondents, common-prior categorical ridge scores 0.13669. Calibrated GA is lower by 0.00705 on Qwen and 0.00278 on Granite. Against the stronger raw-answer control (0.13494), the Qwen difference is $-0.00530$ (crossed interval $[-0.00638, -0.00422]$, conditional Holm $p = .0096$); Granite's $-0.00103$ contrast does not satisfy the combined decision rule ($p = .0656$). Thus a restricted positive Qwen result survives the shared-prior check, whereas a cross-model claim against that control does not. The LLM still uses 120 additional calibration people and pretrained parameters; the ridge fitting count excludes its 120 selection people. These diagnostics neither establish equal-total-budget efficiency nor select an optimal budget on unexposed data.

Under the frozen ridge input curve, the primary Qwen persona corresponds to 10.97 answers (conditional 95% interval $[9.06, 12.92]$) and Granite to 7.51 ($[6.36, 8.84]$). These intervals propagate respondent and nested-order variation after averaging the three persona seeds; they do not propagate fresh optimizer-seed uncertainty. All 84 search curves reach 78.2–80 full-panel equivalents. Their checkpoints describe development efficiency, not a second procedure for selecting on evaluation respondents.

## 25.2 Exploratory complementarity

The selected persona–ridge mixture improves over both components in all 12 conditions under the screen rule. Gains over ridge range from 0.00192 to 0.01158 RPS, and gains over the calibrated persona from 0.00113 to 0.01009 (all conditional Holm $p = .0096$). The supplement shows paired gains and uncertainty for both representations and all 48 contrasts. No new model inference was needed: the experiment reuses saved forecasts, with calibration fitting and mixture selection separated.

The selected mixtures had lower RPS on this exposed cohort; the neutral control separates this observation from optimizer value. On Granite, the optimized mixture has lower mean RPS than the neutral mixture in all six conditions; three pass the 0.001 magnitude threshold as well as the statistical rule (categorical 80/200, raw 80). On Qwen, all six neutral mixtures have slightly lower means, with no significant difference after correction. Population-mixture

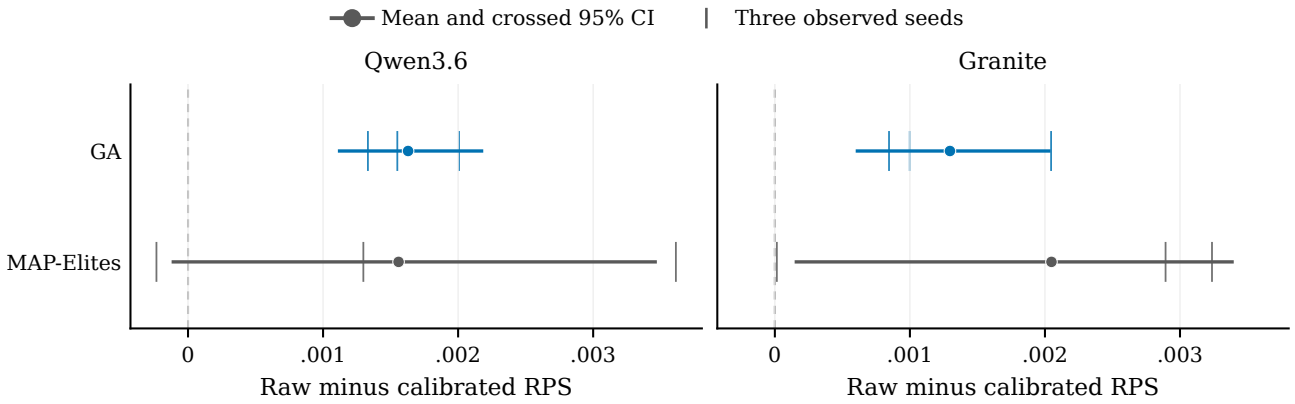



Figure S8: Direct objective effects, post-hoc analysis. Ticks show three paired seed effects; circles and bars show means and crossed 95% intervals. Positive values favor calibrated fitness. All observed GA seeds favor calibrated fitness, but three seeds do not establish generalization over optimizer randomness. Original comparisons remain in the main complete-comparison tables.

selectors choose zero prior weight throughout, reproducing ridge. This linear-pool control is distinct from the earlier prior-anchored intercept refit. The observed advantage of combining predictors spans both tested models, while additional utility from persona search remains model- and budget-specific.

A separately frozen adversarial extension tests prior-anchored ridge and neutral-plus-anchored ridge in all 12 conditions (24-test Holm family). The original hybrid beats anchored ridge throughout, by 0.00161–0.00984 RPS (all $p_{\mathrm{H}} = .0048$). Against the stronger neutral mixture, only Granite categorical/raw 80 clear the magnitude/statistical rule. Qwen loses in all six conditions, five significantly. This practical rival changes both prompt and decoder; it is not a pure optimizer ablation or independent confirmation.

## 26 Detailed saved-forecast protocols

After inspecting the original evaluation, we fixed the following saved-forecast analyses before executing them. They reuse the exposed 600-person cohort and are *post hoc*, not independent confirmation. Within these diagnostics, no prompts are changed and no new LLM predictions are generated. The original 110-test analysis is retained.

Representation controls compare the exact 175 one-hot profile features visible in prompts with 35 continuous pre-discretization values and raw history. All use Train8000 and Select120; original losses reproduce within $10^{-10}$. Separate Holm families cover 33 representation contrasts and nine objective/optimizer contrasts. Rank diagnostics retain the first complete evaluation per genotype in all 24 GA/MAP search pools.

Budget diagnostics fit categorical and raw-history ridge at 80, 200, 320, 1,000 and 8,000 people and exact Search80. Ten fixed nested subsets per smaller budget share Select120 penalty choice. A subsequent prior-anchored control replaces the subset target mean with the Train8000 prior. Both curves reproduce full-data losses. Each has a separate 48-test family; intervals condition on fits and resample respondents and observed LLM runs. Subset ranges are not confidence intervals. Full specifications and the adaptive exposure history are in the supplement.

**Exploratory forecast combination.** A final fixed screen tests complementarity using a linear pool of ordinal CDFs Gneiting and Ranjan [2013], $H = (1 - w)R + wP$, where $R$ is ridge and $P$ a calibrated-GA persona. Both models, both ridge representations and fitting sizes 80, 200 and 320 are included. Three fixed nested training subsets cross all three saved optimizer runs. Ridge penalties use Select. The first 60 Calibrate people fit the persona scale and center; the remaining 60 select $w \in \{0, .05, \ldots, 1\}$ by RPS. Parameters are fixed before evaluation. Controls are ridge, persona, neutral-persona plus ridge, and population-prior plus ridge; each mixture tunes its own weight on the same 60 people. A separate 48-test Holm family and 5,000 respondent-bootstrap draws condition on the nine fixed fit/run combinations. The screen requires mean gain at least 0.001, a positive gain interval and adjusted $p < .05$. All 12 conditions are reported. This adaptive analysis reuses the exposed cohort and the 8,000-person prior; low ridge fitting counts do not mean low total supervision.

## 27 Detailed representation contrasts

The additional categorical-profile ridge scores 0.11384, compared with 0.11558 for continuous-profile ridge and 0.10955 for raw-history ridge (Supplementary Figure S3). Its deficit to raw-history ridge is 0.00428 (paired respondent interval $[0.00378, 0.00478]$), while every LLM arm is 0.01581–0.02388 worse than categorical ridge; all 28 intervals are positive with Holm $p < .01$ in the 33-test family. Neutral personas also lose by 0.01625 on Qwen and 0.02273 on Granite. The profile retains information usable by a simple predictor even when persona search adds little. Ridge fits 8,000 person-level labelled mappings; the LLM pipeline uses those labels for item priors, while prompt optimization uses 80 search and 120 selection people. Thus person-level supervision budgets are not matched. This does not equate their learning procedures or give a causal compression/readout decomposition.

Relative to the empirical prior, categorical ridge retains $(0.15174 - 0.11384)/(0.15174 - 0.10955) = 89.8\%$ of raw-history ridge's RPS improvement. This is an observed prediction-improvement ratio, not a fraction of human information. Calibrated GA and MAP-Elites recover only 2.1% and 1.6% of their base-to-raw-ridge gaps on Qwen and 9.8% and 9.9% on Granite. These descriptive ratios have no additional significance claim.

## References


Tilmann Gneiting and Roopesh Ranjan. Combining predictive distributions. *Electronic Journal of Statistics*, 7: 1747–1782, 2013. doi: 10.1214/13-EJS823. URL `https://doi.org/10.1214/13-EJS823`.

John A. Johnson. Measuring thirty facets of the five factor model with a 120-item public domain inventory: Development of the IPIP-NEO-120. *Journal of Research in Personality*, 51:78–89, 2014. doi: 10.1016/j.jrp.2014.05.003. URL `https://doi.org/10.1016/j.jrp.2014.05.003`.